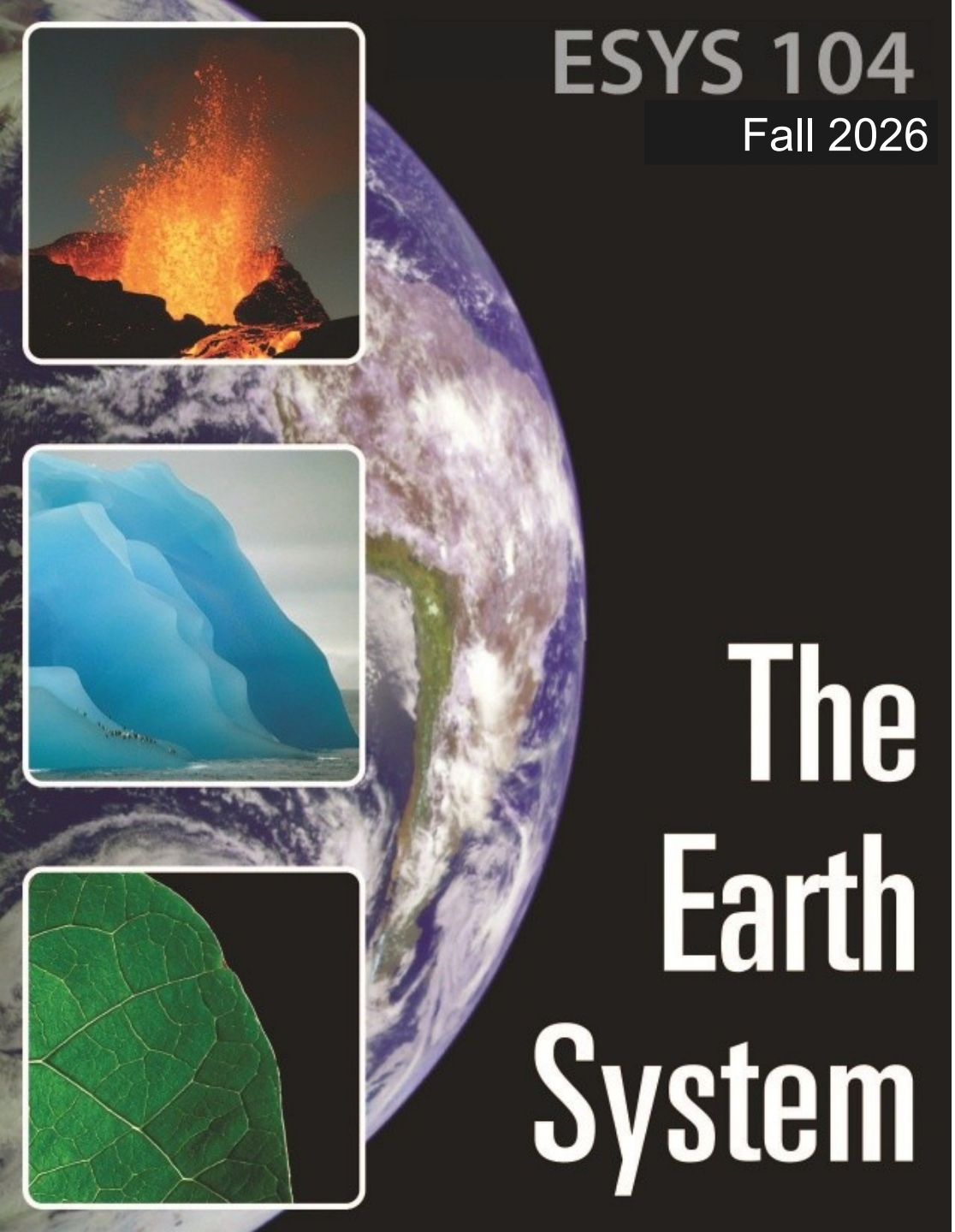
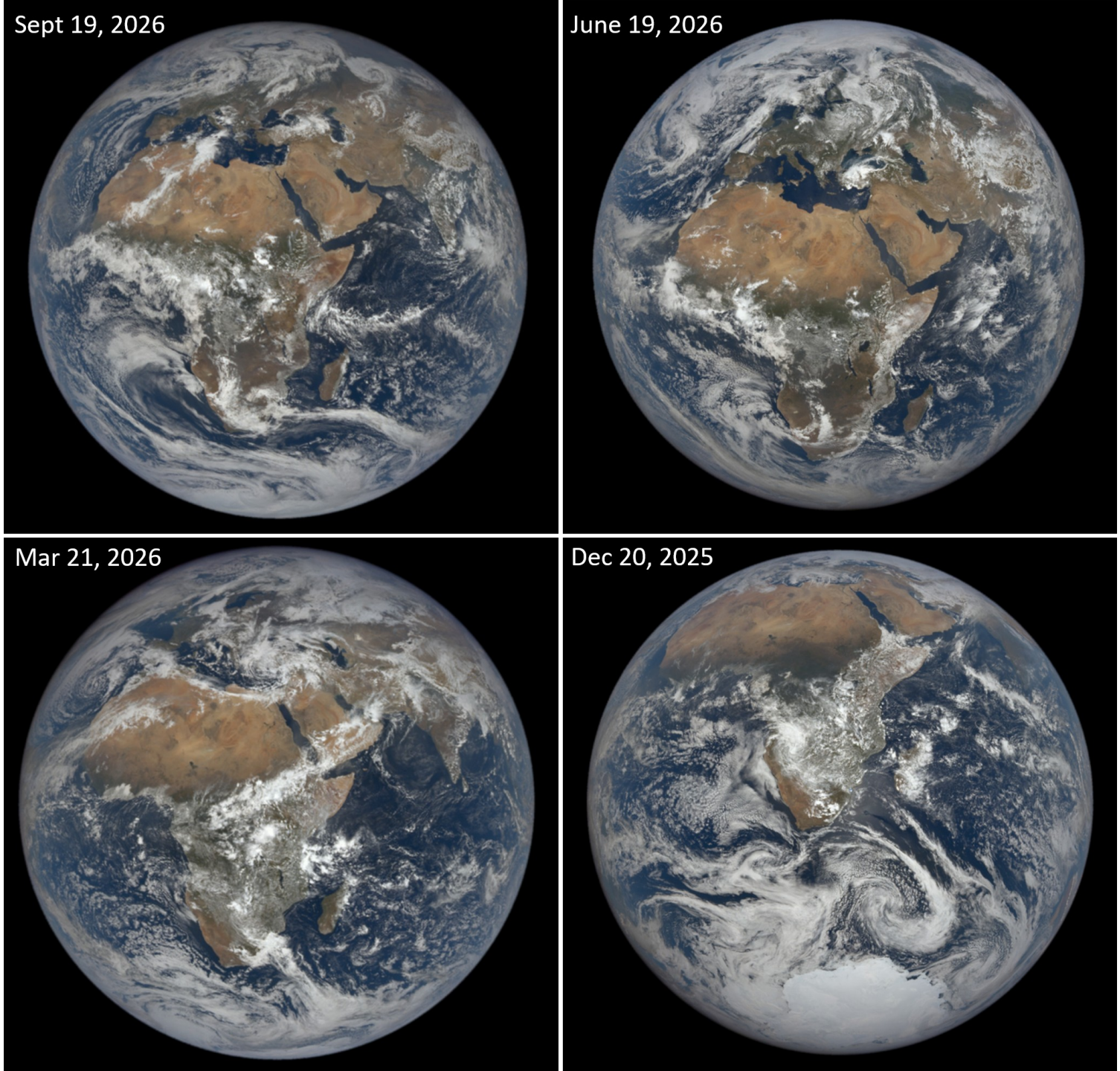


#7: Atmosphere circulation, ocean circulation

Assignment #1 due Monday



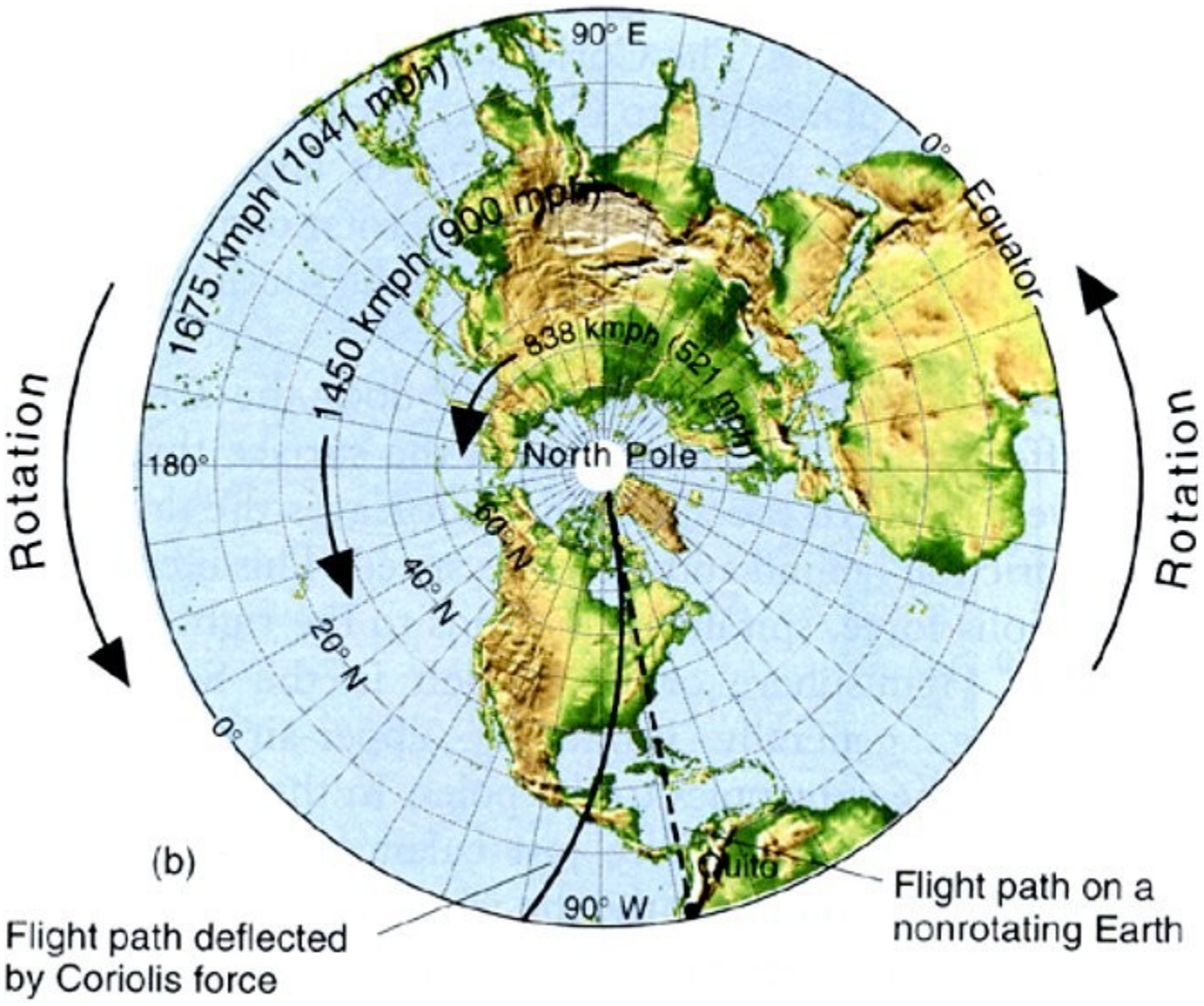
DISCOVR

Victoria BC: 27Jan25
Radiative Heat Transfer



Ed Wiebe

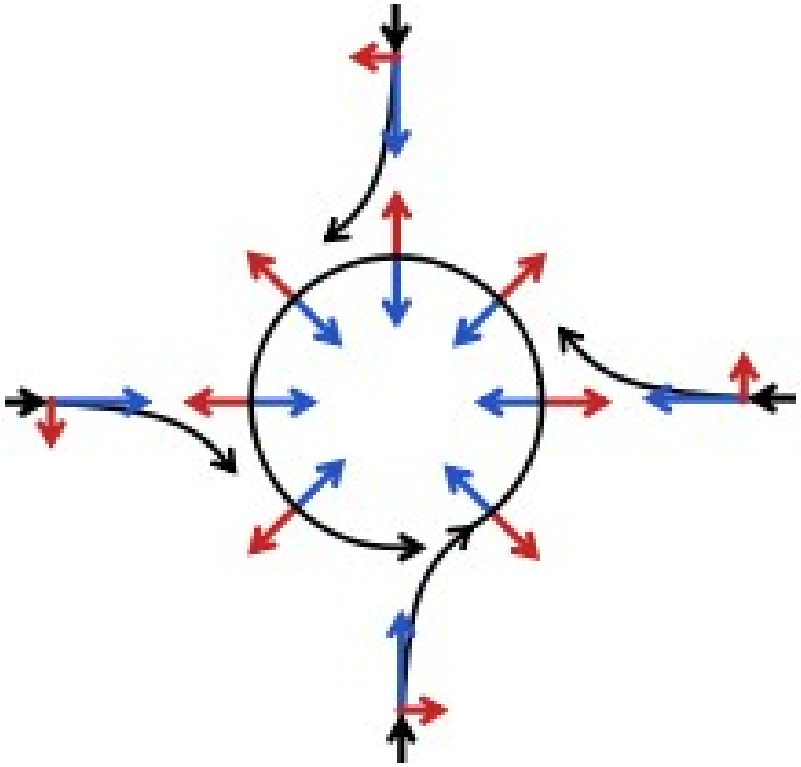
Further Twist: The Coriolis Effect



Because we are on a rotating Earth, straight paths appear curved to us, as if a “force” was pulling on moving objects. Trajectories curve to the right in the Northern Hemisphere, and to the left in the Southern.

Good thorough explanation of Coriolis effect: http://www.dvandom.com/coriolis/

Low Pressure – CounterClockWise (Northern Hemisphere)



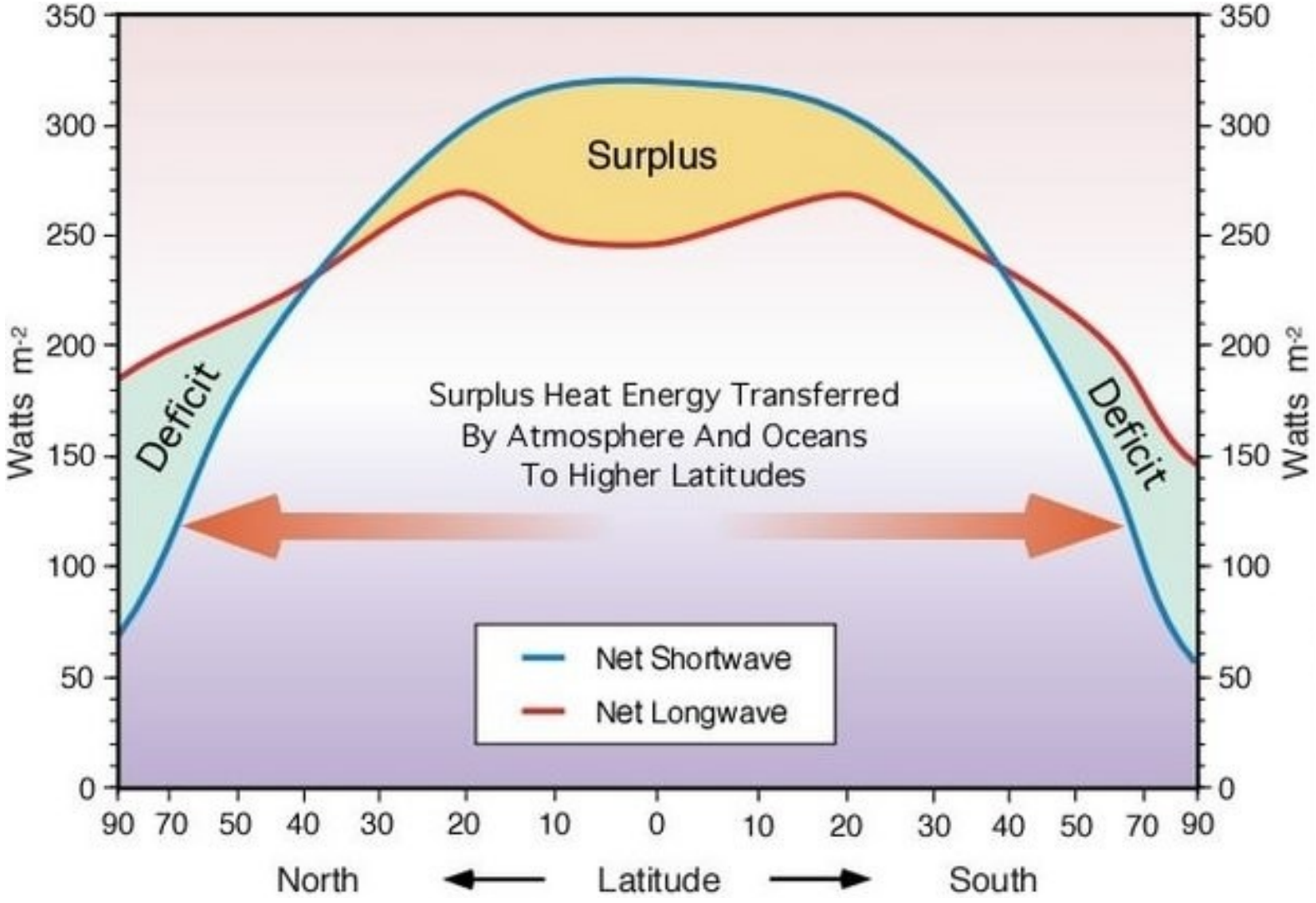


Blue: winds blowing from high to low pressure

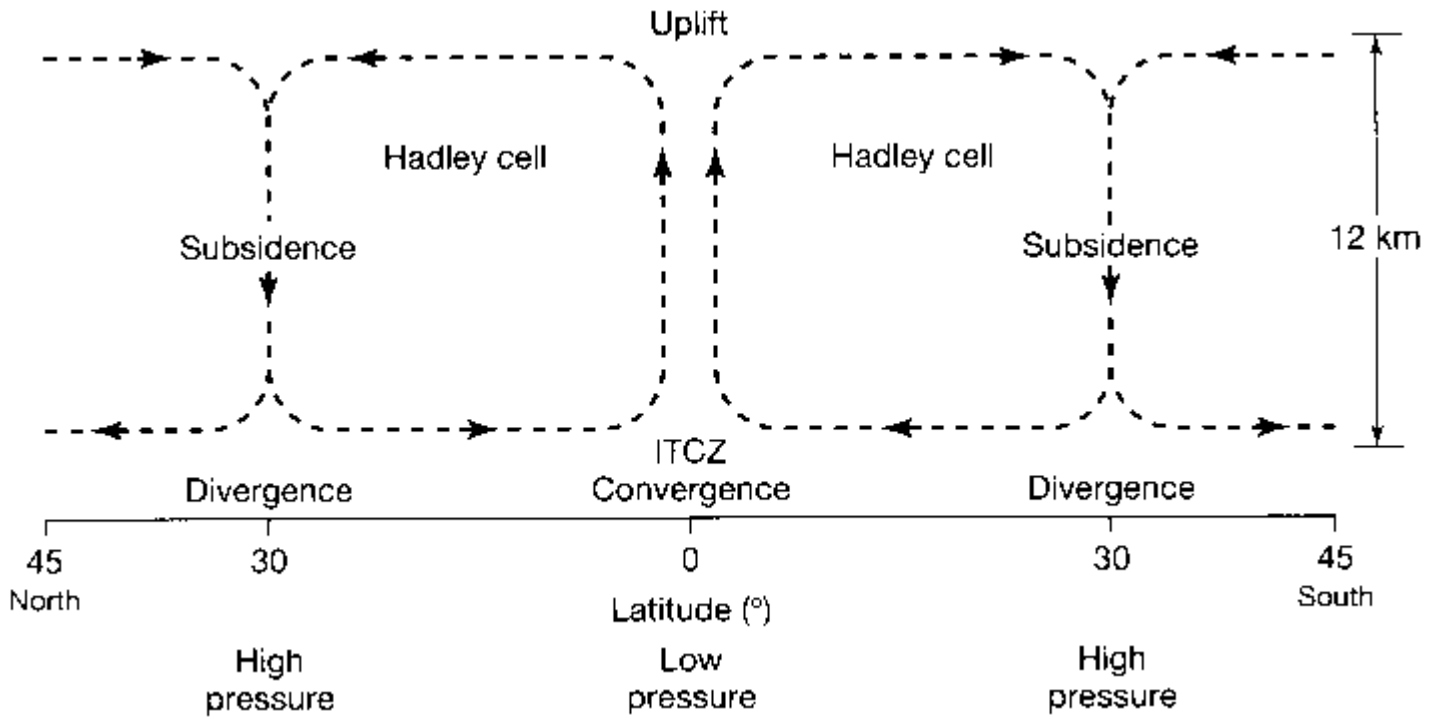
Red: Coriolis effect curving the winds to the right

Black: the path of the winds as the forces balance

Energy Balance of the Earth
Top of the Atmosphere



General Circulation – Tropics



- Warm moist equatorial air rises and moves towards the poles (Hadley cell), leading to cloud formation;
- There is sinking motion at about 30° latitude in both hemispheres → Sunny and dry areas

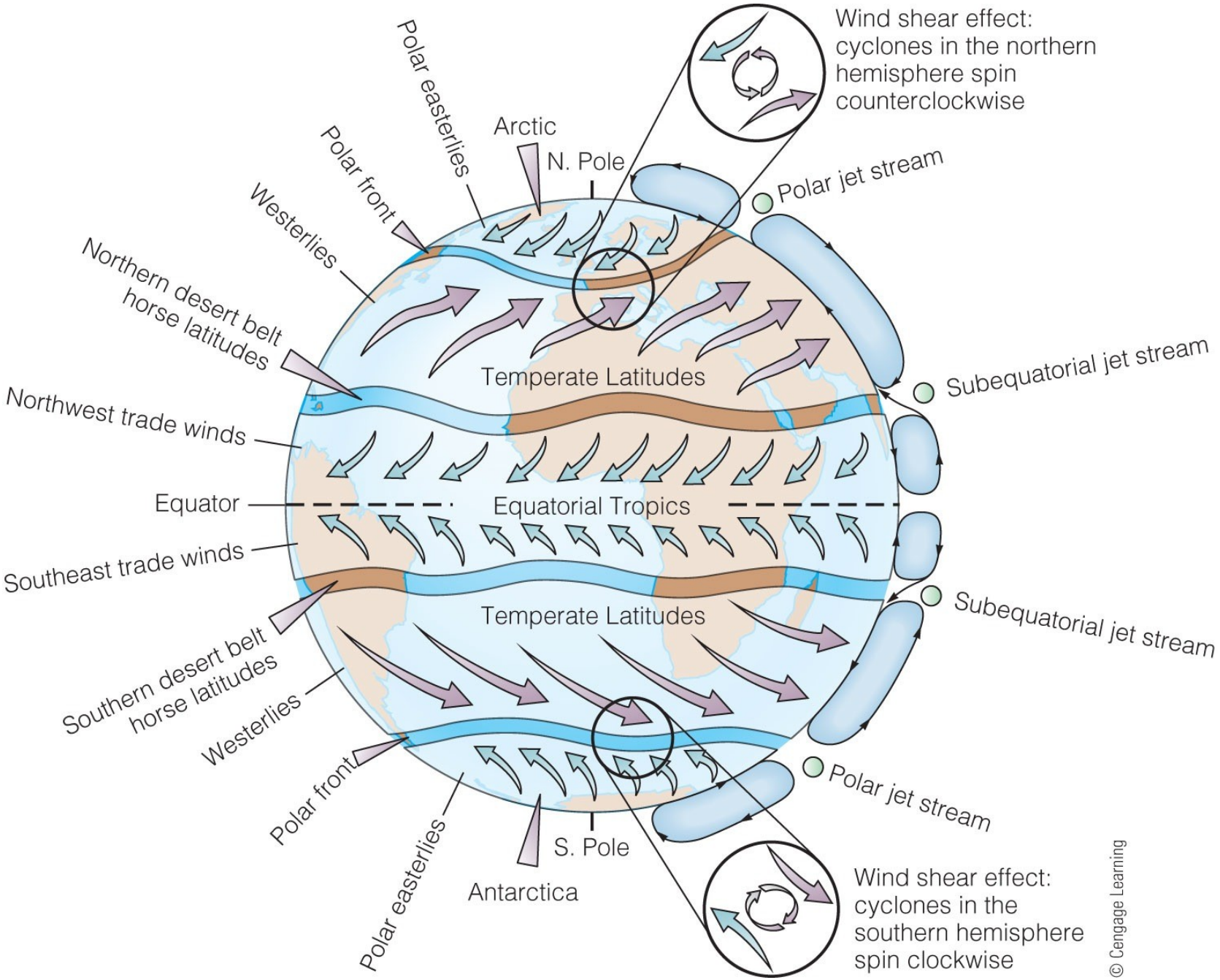
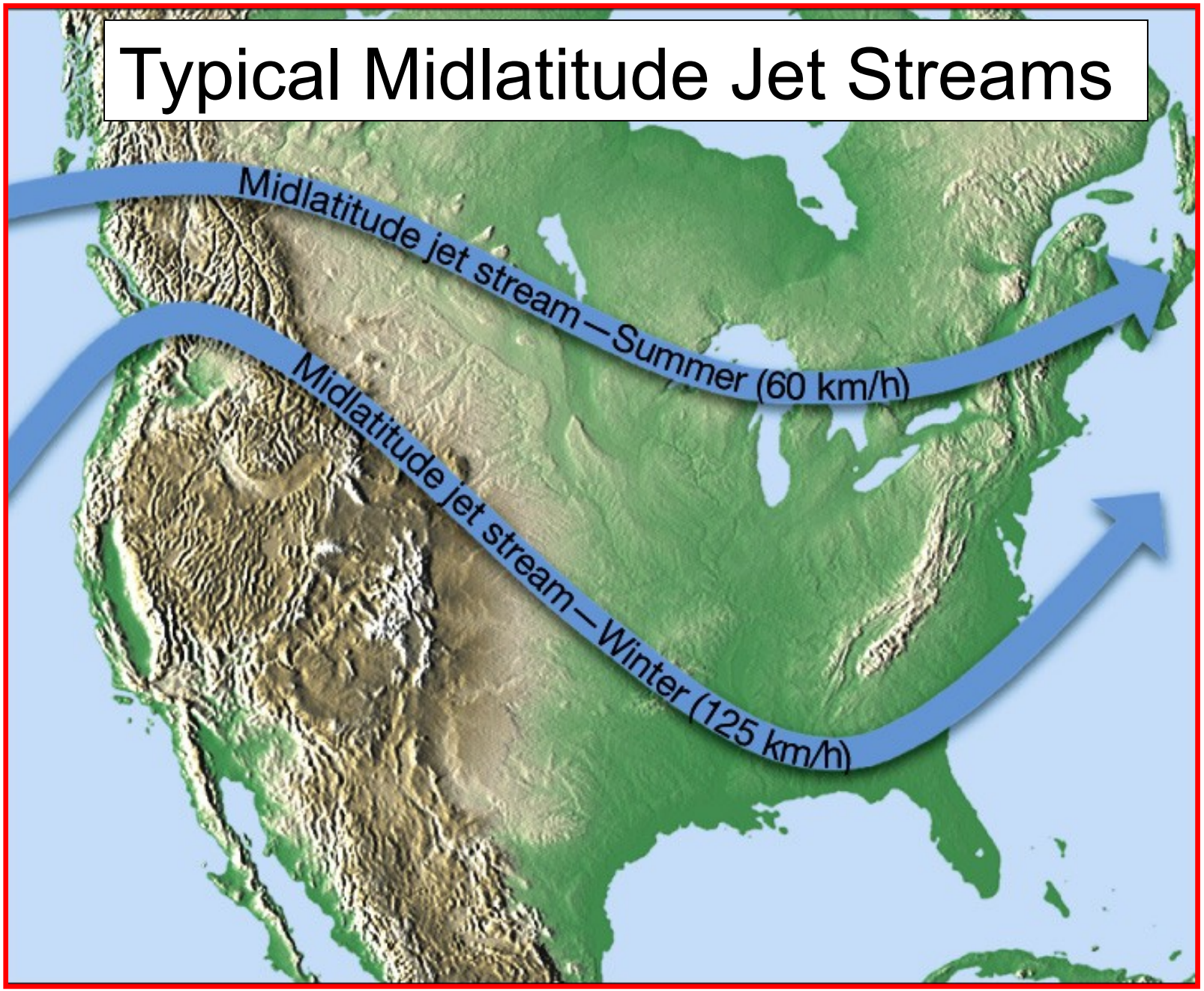
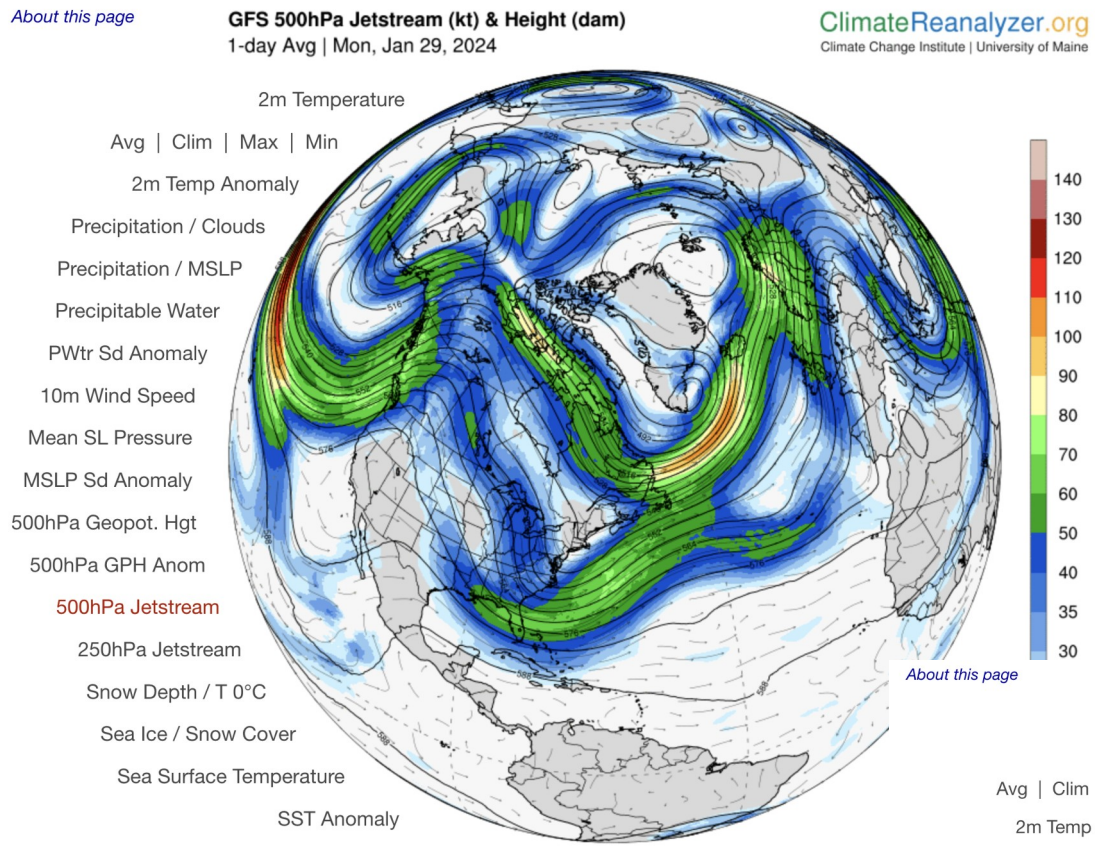
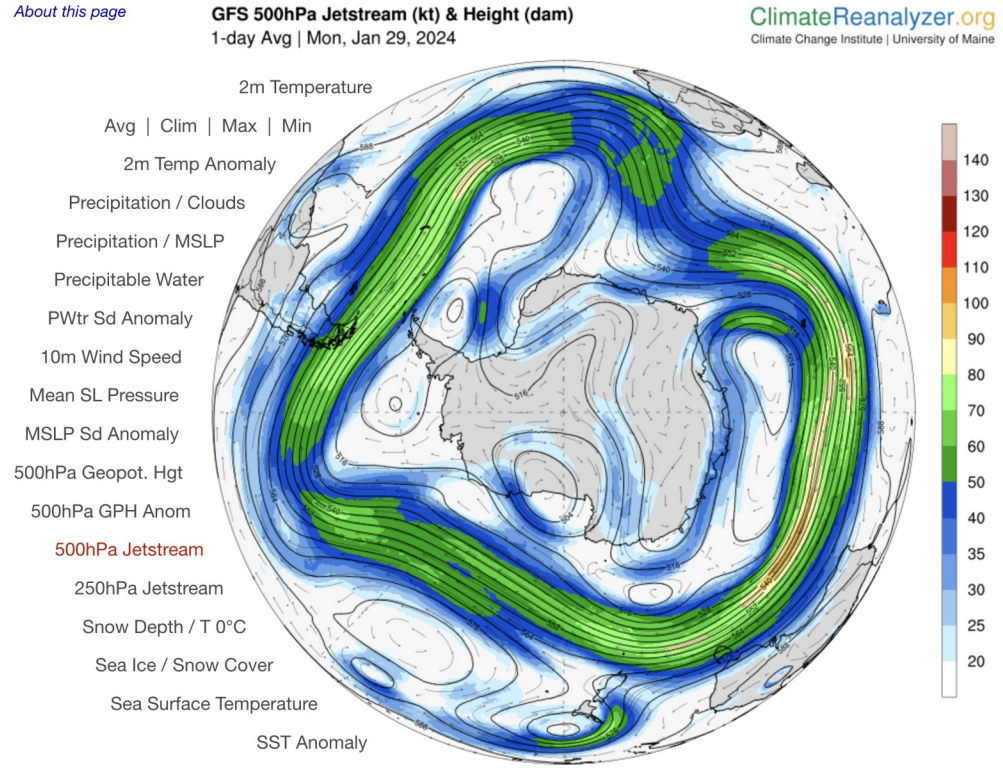


Figure 2-5 p27

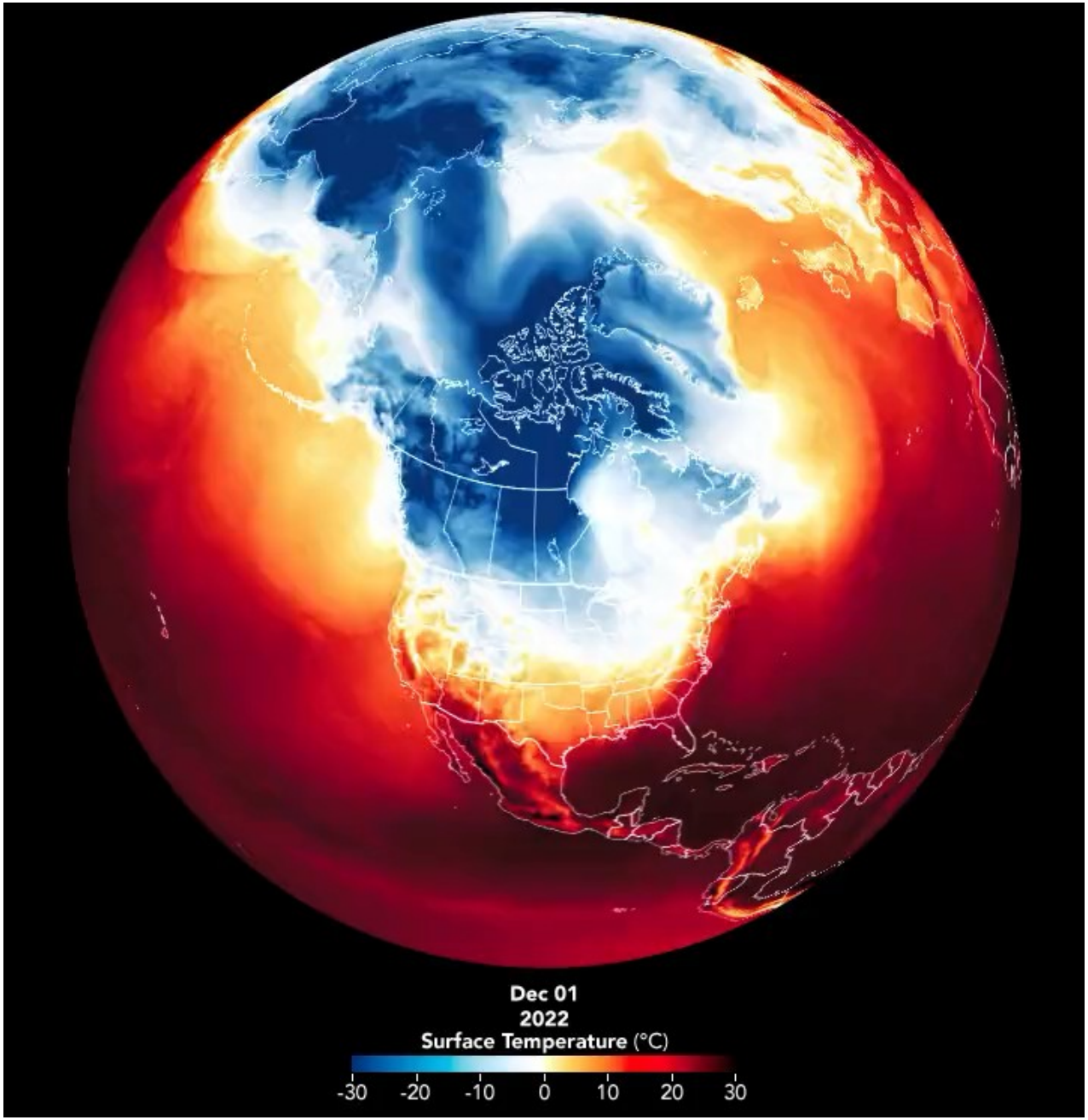

Typical Midlatitude Jet Streams
Midlatitude jet stream—Summer (60 km/h)
Midlatitude jet stream—Winter (125 km/h)



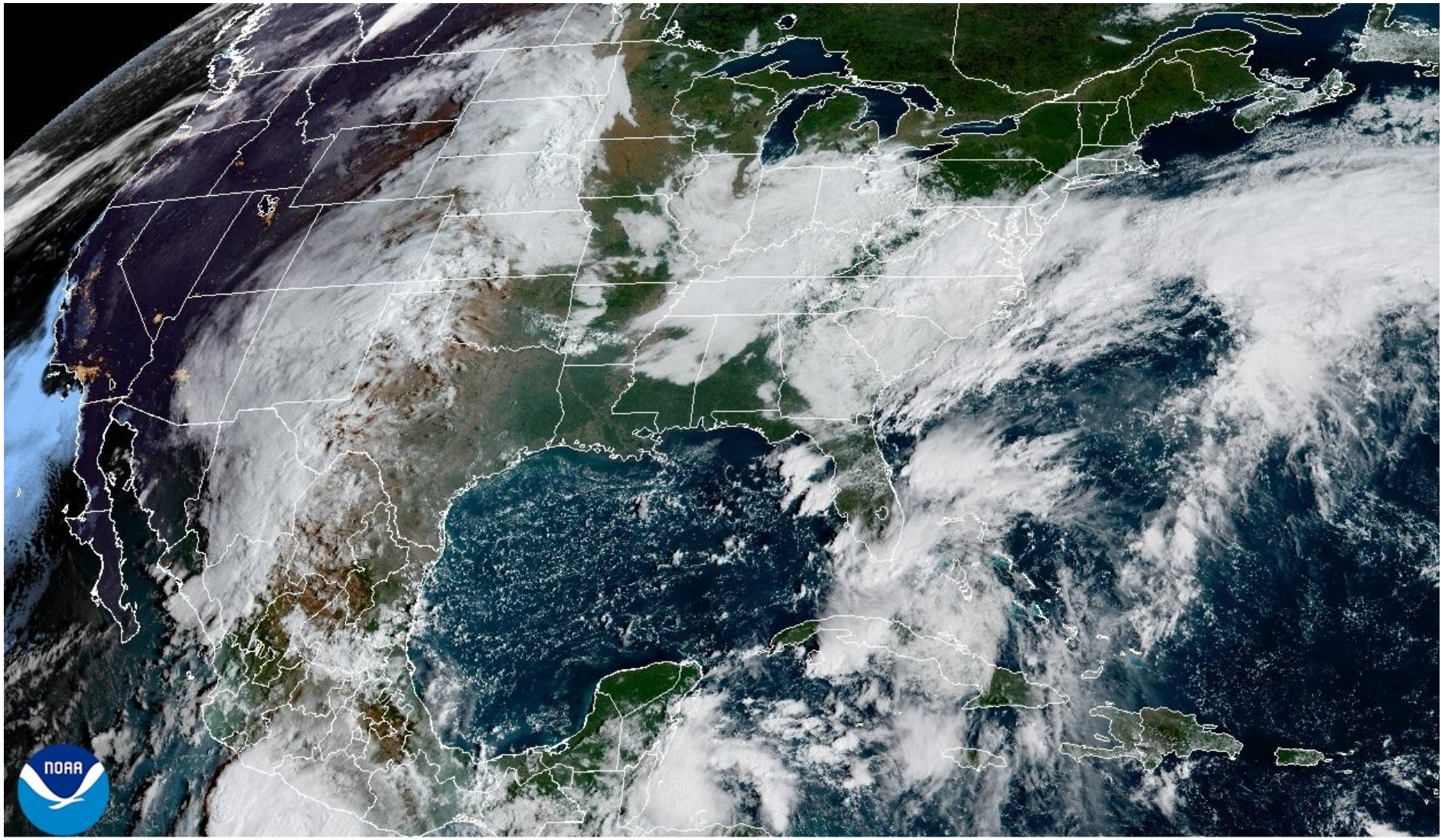
rs



South polar cell anchored by strong temperature gradient

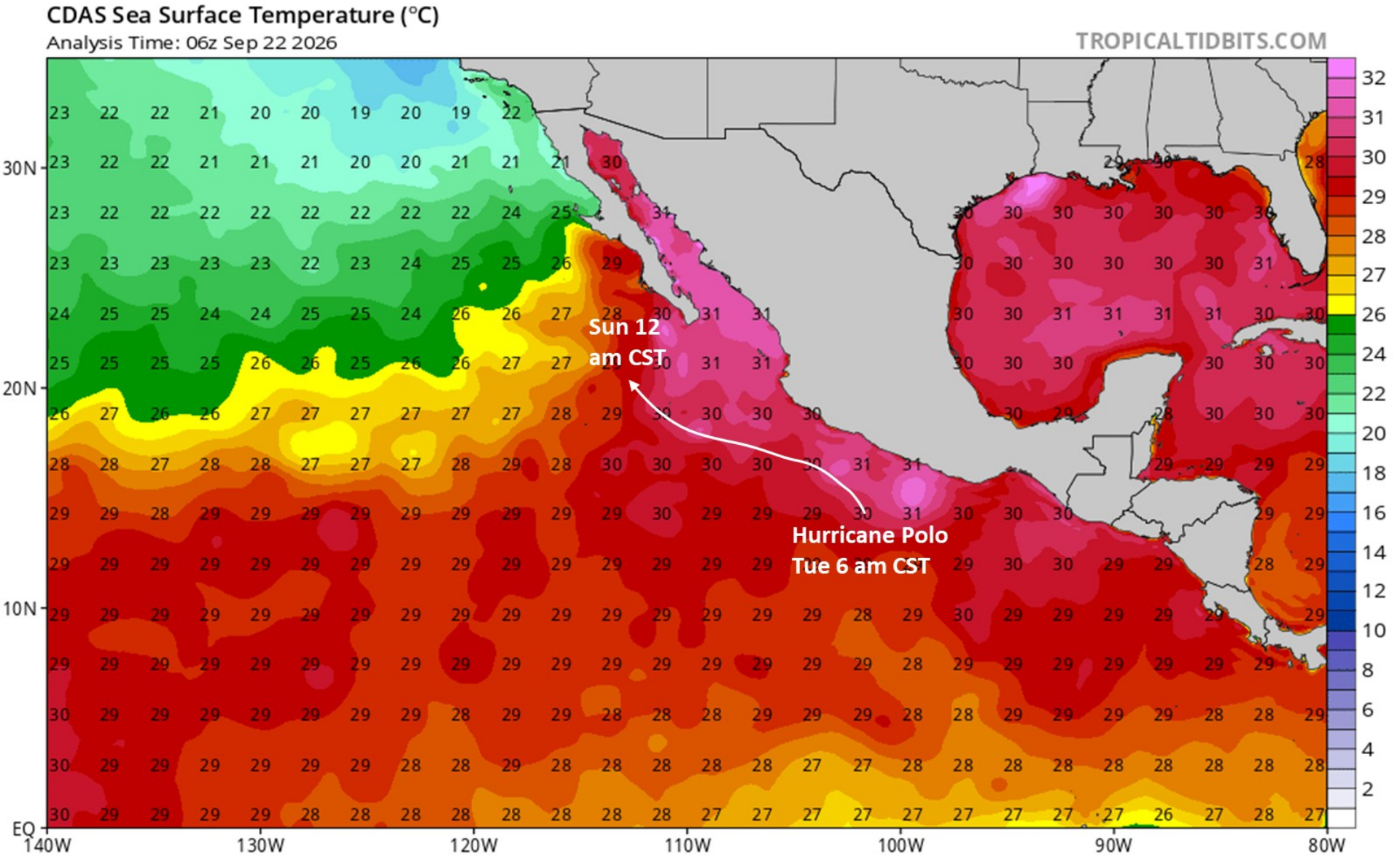

Dec 01
2022
Surface Temperature (°C)
-30
-20
-10
0
10
20
30

Ocean Moisture



23 Sep 2026 13:56Z - NOAA/NESDIS/STAR - GOES-19 - GEOCOLOR Composite

Hurricane Polo

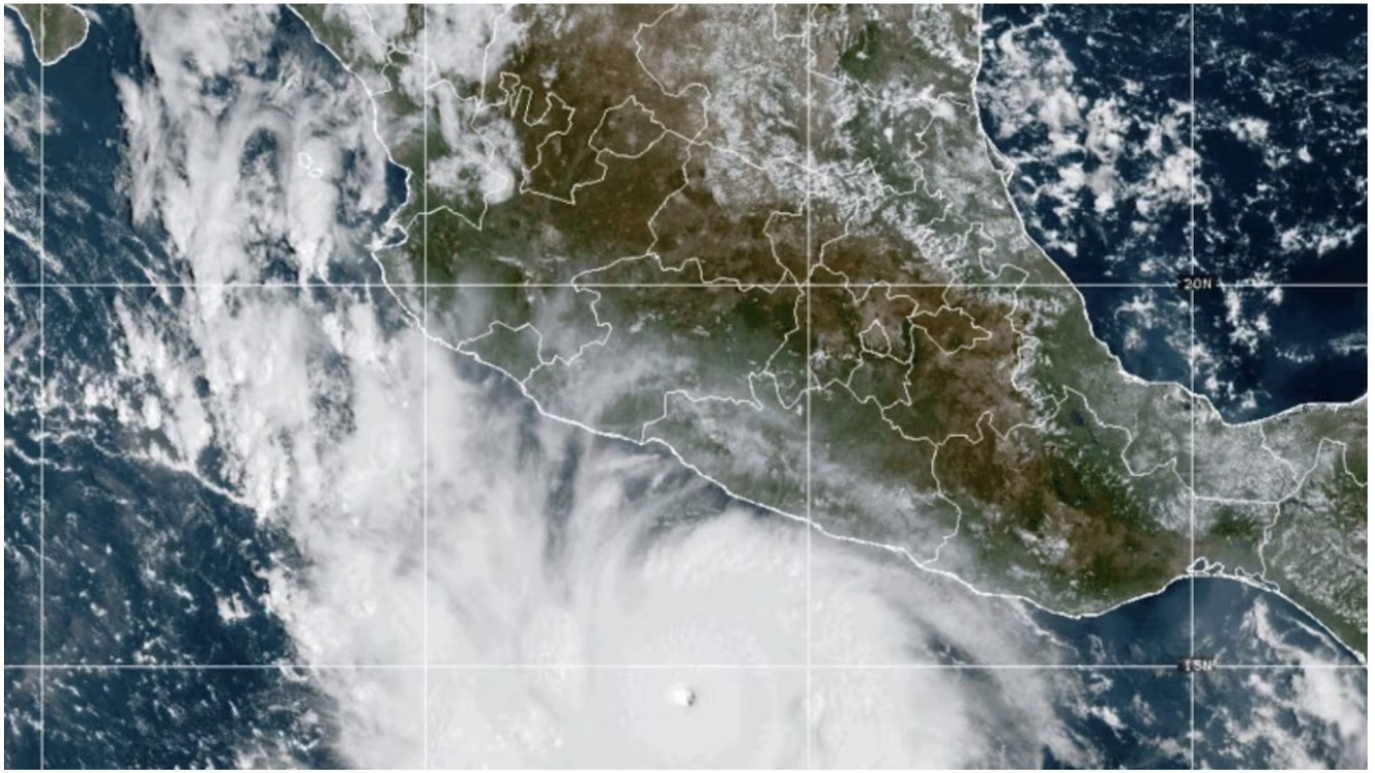


Hurricane Polo in the Pacific Ocean considered 'one of the strongest storms ever'

Polo not expected to make landfall, but its gathering strength is 'truly remarkable,' storm researcher says

The Associated Press · Posted: Sep 23, 2026 7:07 AM EDT | Last Updated: 2 hours ago

Listen to this article
Estimated 5 minutes



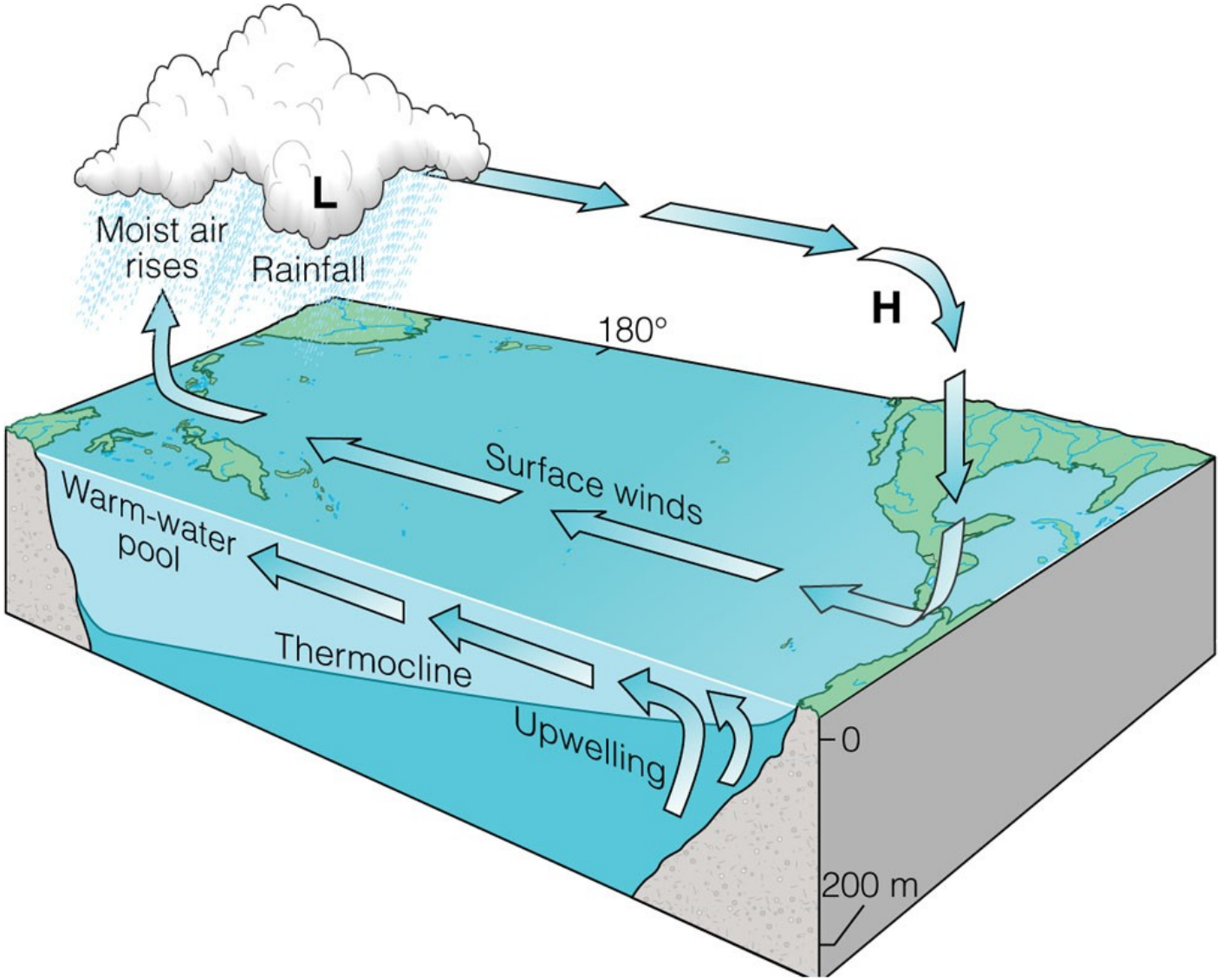
This photo provided by National Oceanic and Atmospheric Administration shows Hurricane Polo strengthening in the Pacific Ocean off the coast of Mexico on Tuesday. (NOAA/The Associated Press)

Oceans



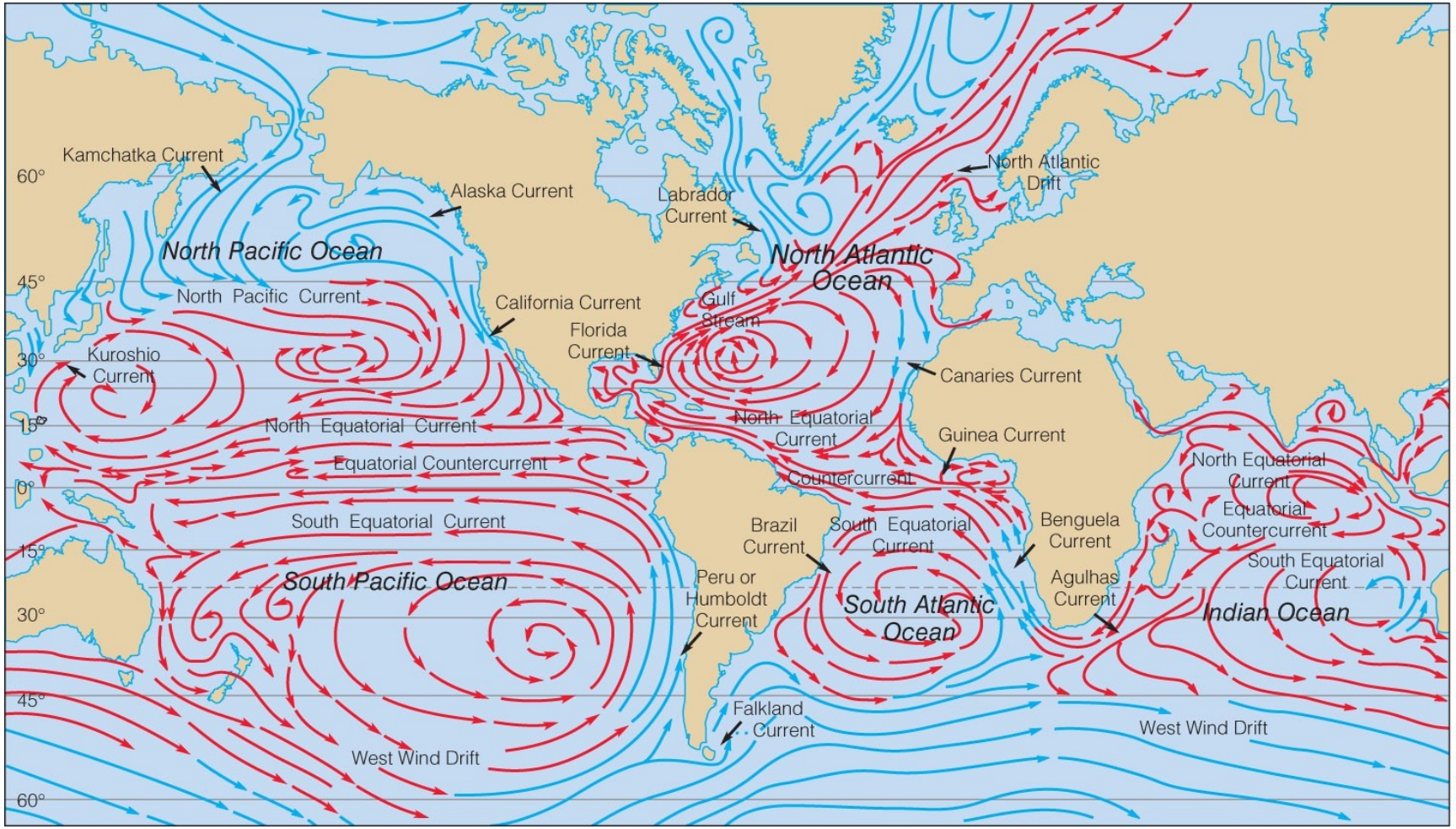
Oceans and seas cover 72% percent of the surface of the Earth and contain 97% of the Earth's surface water.

Surface winds Drive Ocean Currents



© 2005 Brooks/Cole - Thomson

Surface Water Circulation



Gyres are formed through winds and friction, Earth's rotation, with the location of land masses blocking the circulation

Ocean Composition

Salt Content of the Earth's Oceans

Salt Ion	*Grams per Kilogram (g/kg) of Ion in Seawater*	*Ion by Weight (%)*
Chloride (Cl^-)	18.980	55.04
Sodium (Na^+)	10.556	30.61
Sulfate (SO_4^{2-})	2.649	7.68
Magnesium (Mg^{2+})	1.272	3.69
Calcium (Ca^{2+})	0.400	1.16
Potassium (K^+)	0.380	1.10
Bicarbonate (HCO_3^-)	0.140	0.41
Bromide (Br^-)	0.065	0.19
Boric acid (H_3BO_3)	0.026	0.07
Strontium (Sr^{2+})	0.013	0.04
Flouride (F^-)	0.001	0.00
Total	34.482	99.99

Source: PINET, P. R., *Oceanography*, St. Paul, MN: West Publishing Co., 1992.

Grams per kilogram: parts per thousand or parts per mil (‰)

Crystallizing Minerals 'Unmix' solutions

Seawater solutes

Element	Concentration (by mass)
Chlorine	1.94%
Sodium	1.08%
Magnesium	0.129%
Sulfur	0.091%
Calcium	0.04%
Potassium	0.04%
Bromine	0.0067%
Carbon	0.0028%

(mostly carbonate and bicarbonate)

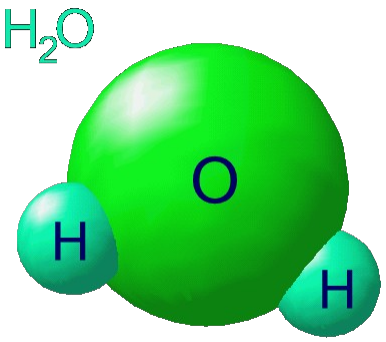
Seawater as an example

NCAR



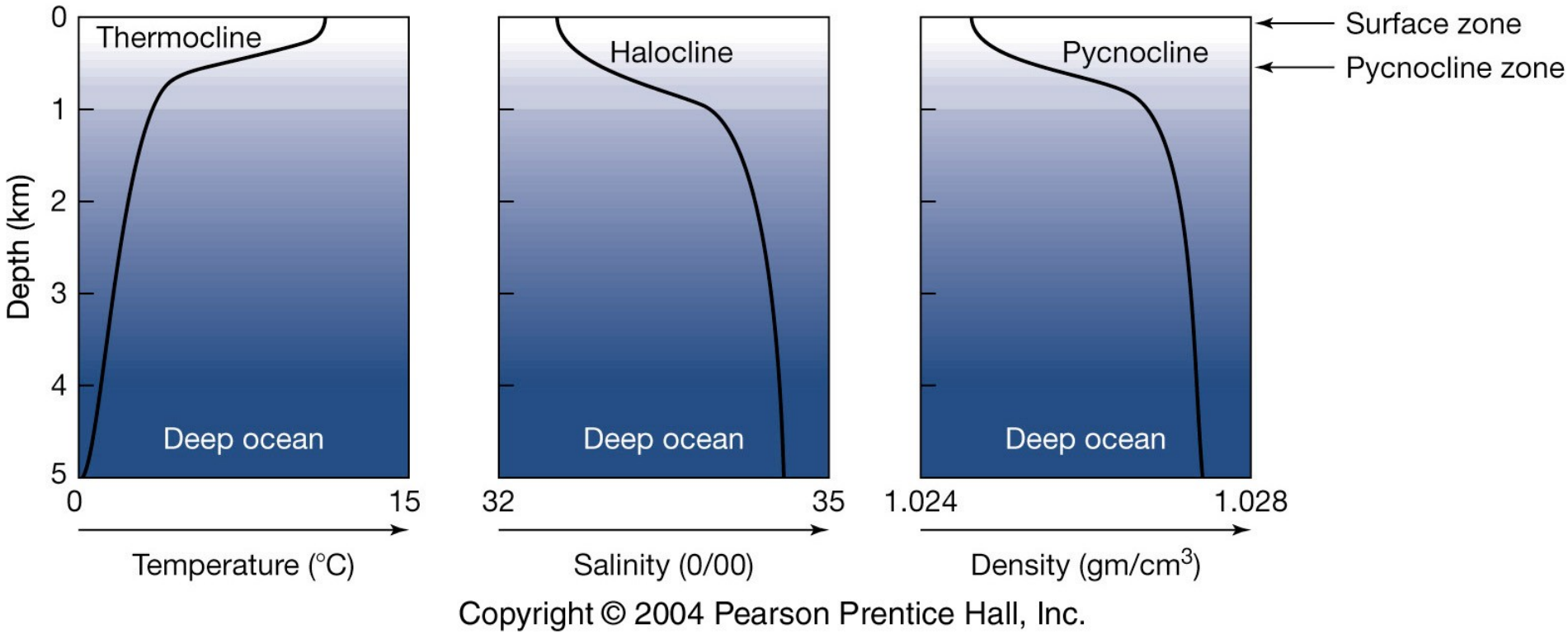
Sea Ice:
almost pure water (H_2O)

Some Peculiar Properties of Water



- Great solvent;
- Melting point: 0°C; lower for salty water (typically around −4°C in oceans);
- Pure water is densest at 4°C (and around −2°C for salty ocean water);
- Solid water (ice) is less dense than liquid water (920 vs 1000 kg/m^3).

Heat and Salt Exchange



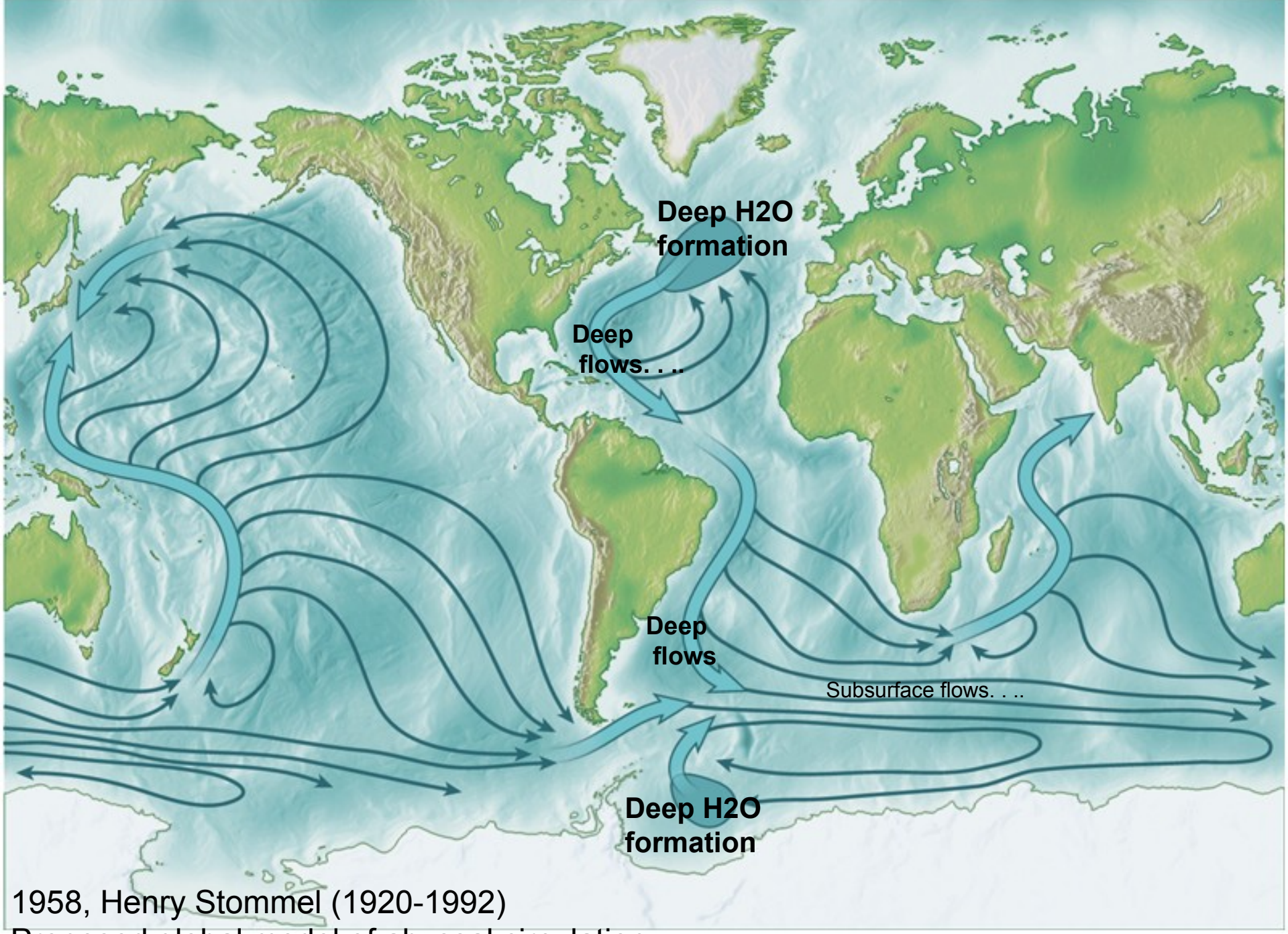
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Sea surface salinity is changed by precipitation, evaporation, runoff and freezing/melting of sea-ice.

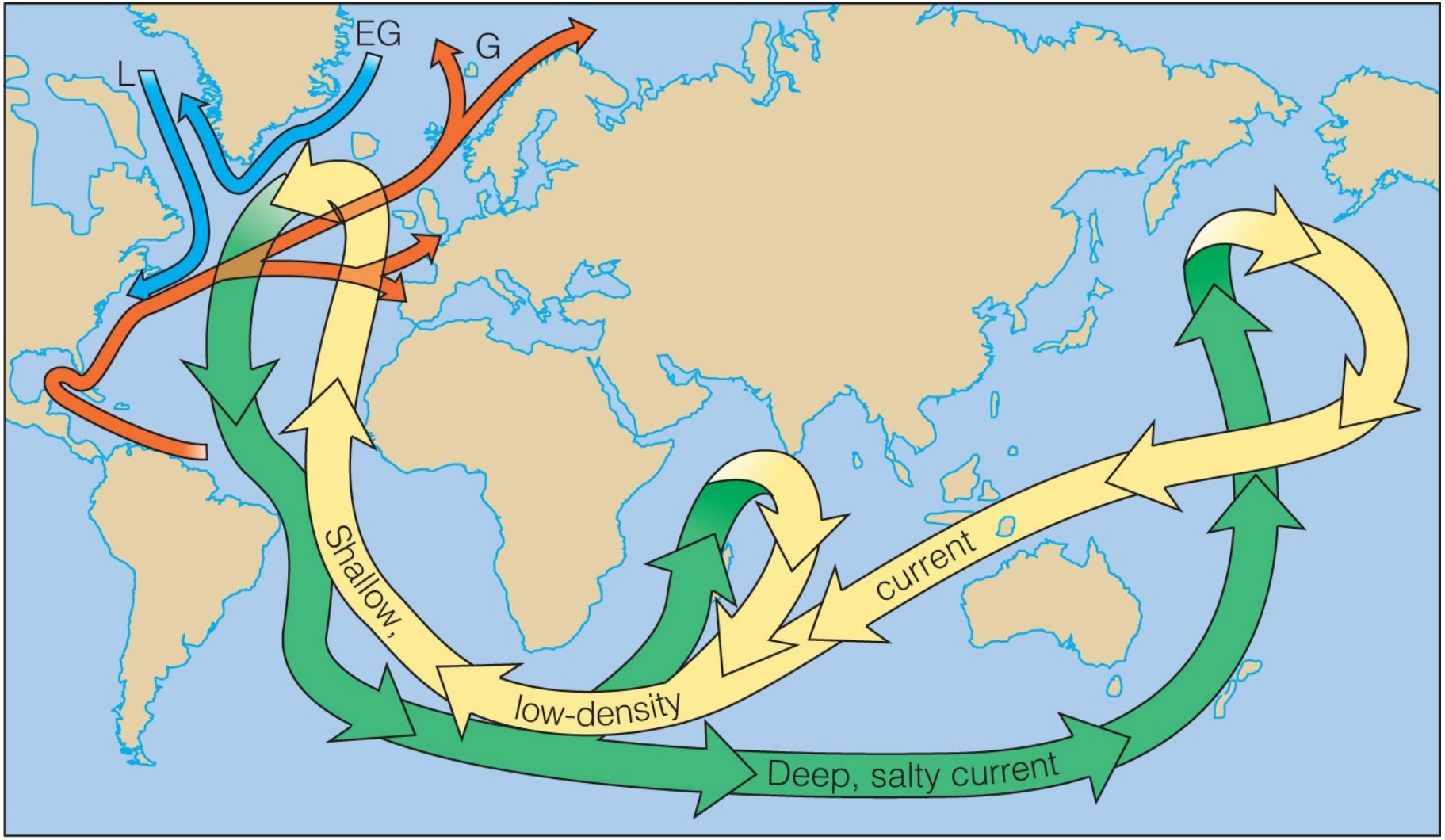
Causes of Deep Ocean Circulation

- Seawater contains dissolved salts (concentration varies with location);

- Changes in temperature (thermo-) and salinity (haline) drive the deep currents (thermohaline circulation).



1958, Henry Stommel (1920-1992)
Proposed global model of abyssal circulation

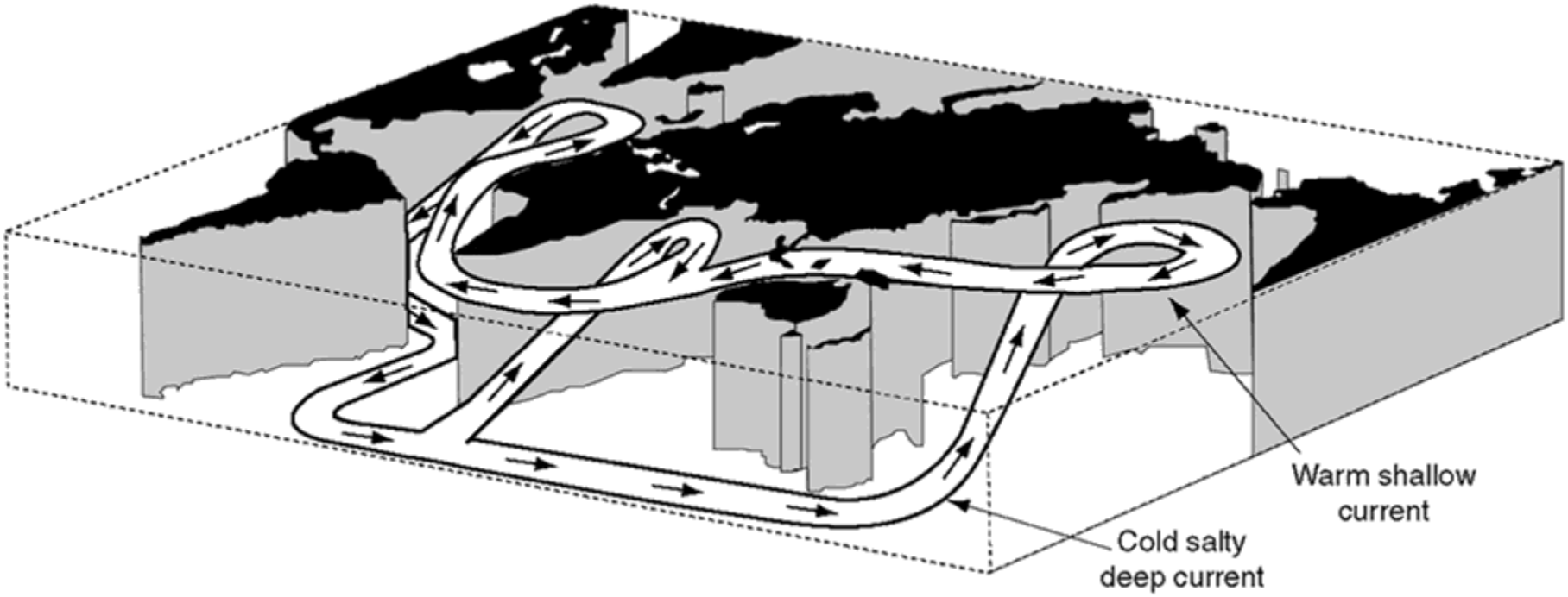


Data from Broecker, W. S. 1995, November. Chaotic climate. *Scientific American* 273[5].)

Figure 2-7 p30

Conveyor Belt

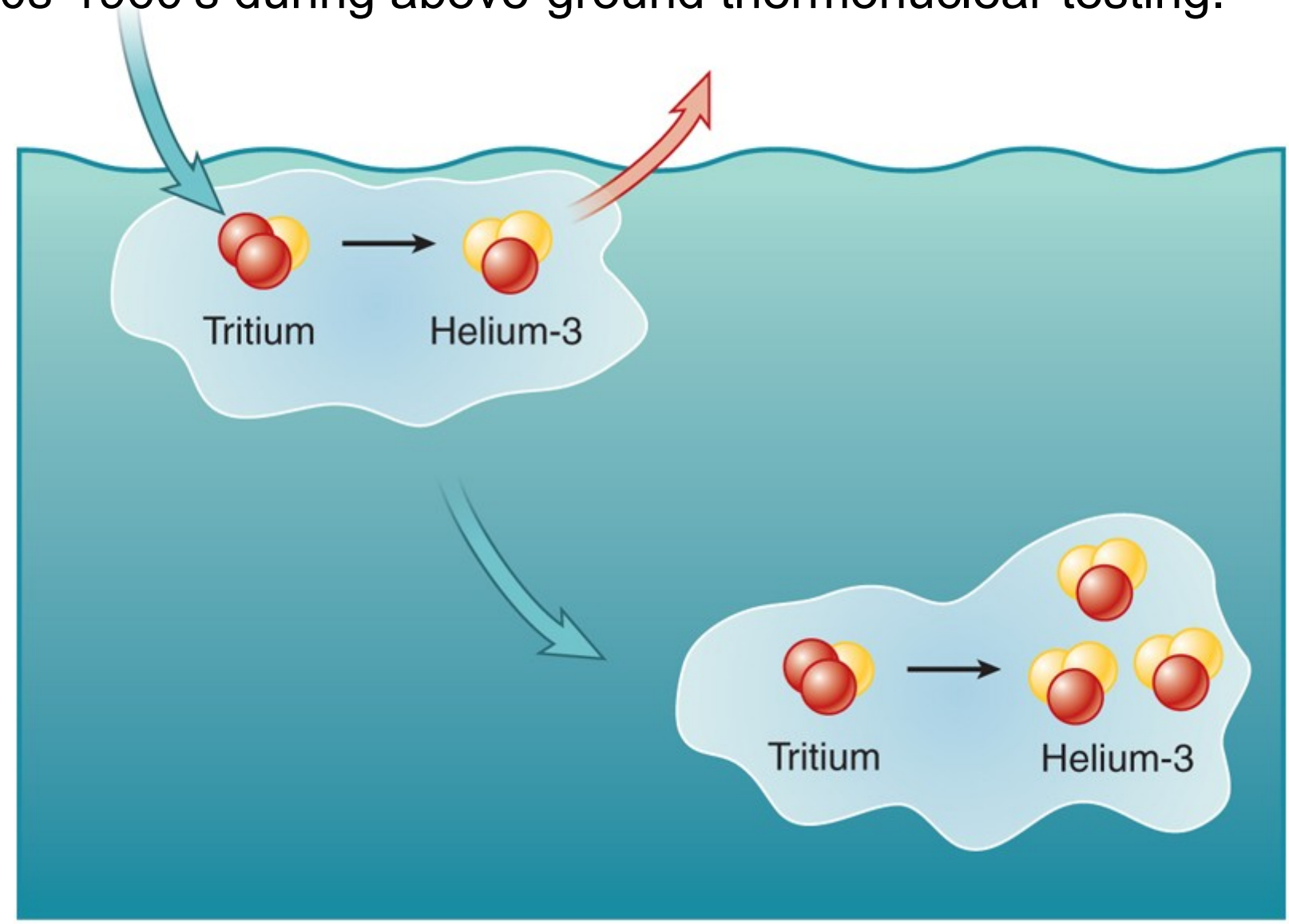
To be thought of as a “mean circulation” with lots of fluctuations around



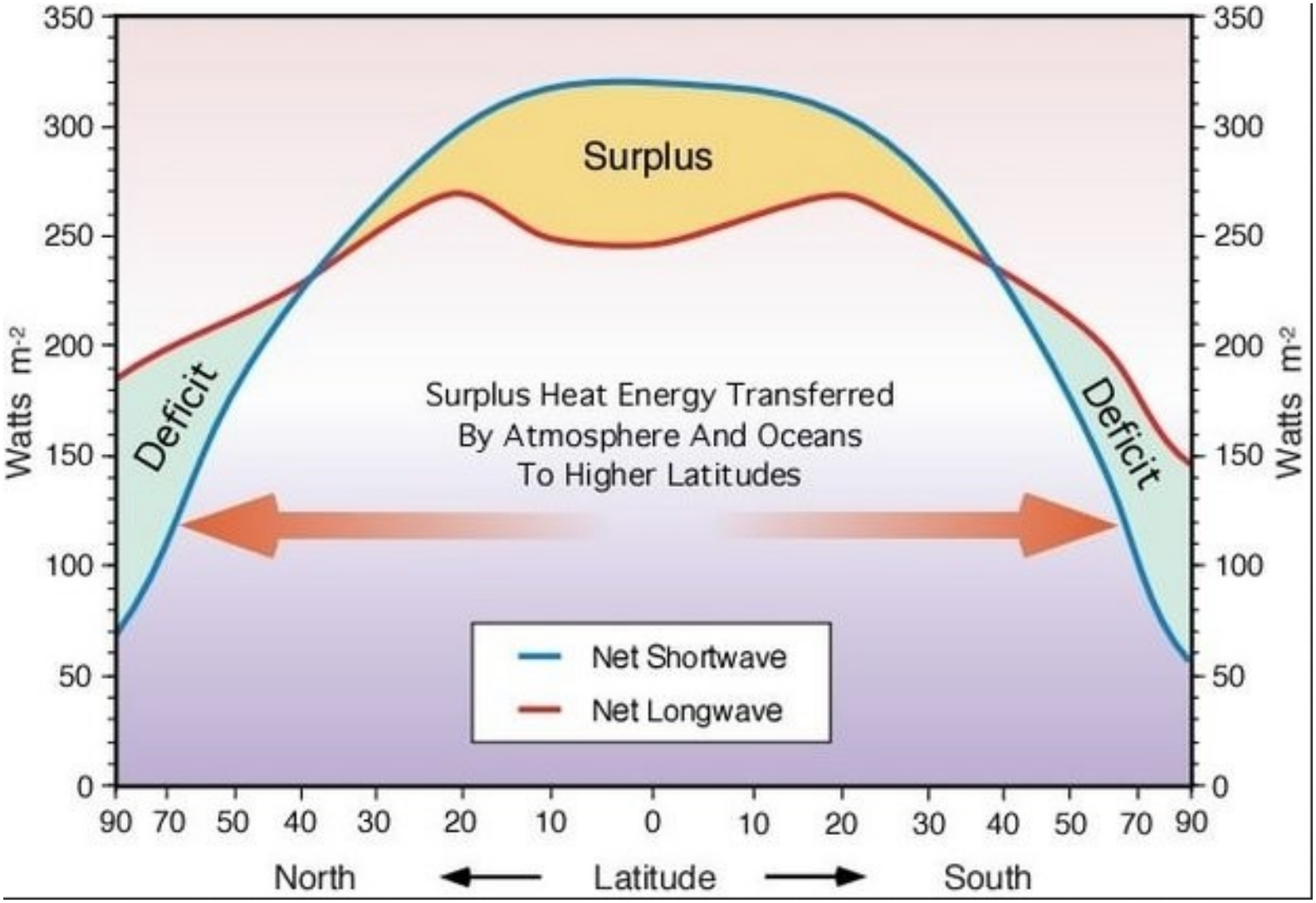
- Dense water forms and sinks in the North Atlantic, setting up conveyor (about 1000-2000 year recycling time scale);
- Paleoclimate records suggest the conveyor has changed in the past due to freshwater input from ice cap melting.

Tracers of Deep Water Circulation

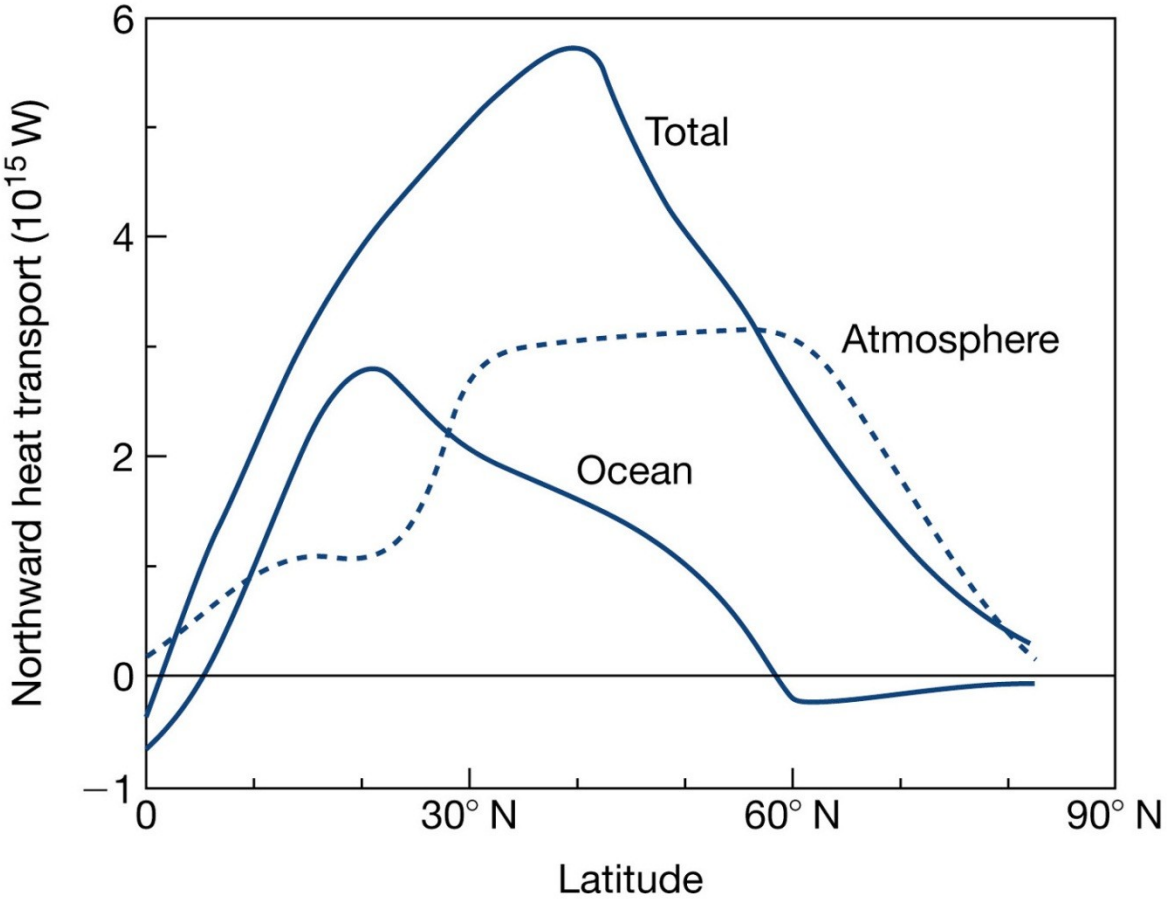
Tritium is radioactive (12.7 years half-life); only produced in the 1950s-1960's during above-ground thermonuclear testing.



Energy Balance of the Earth
Top of the Atmosphere



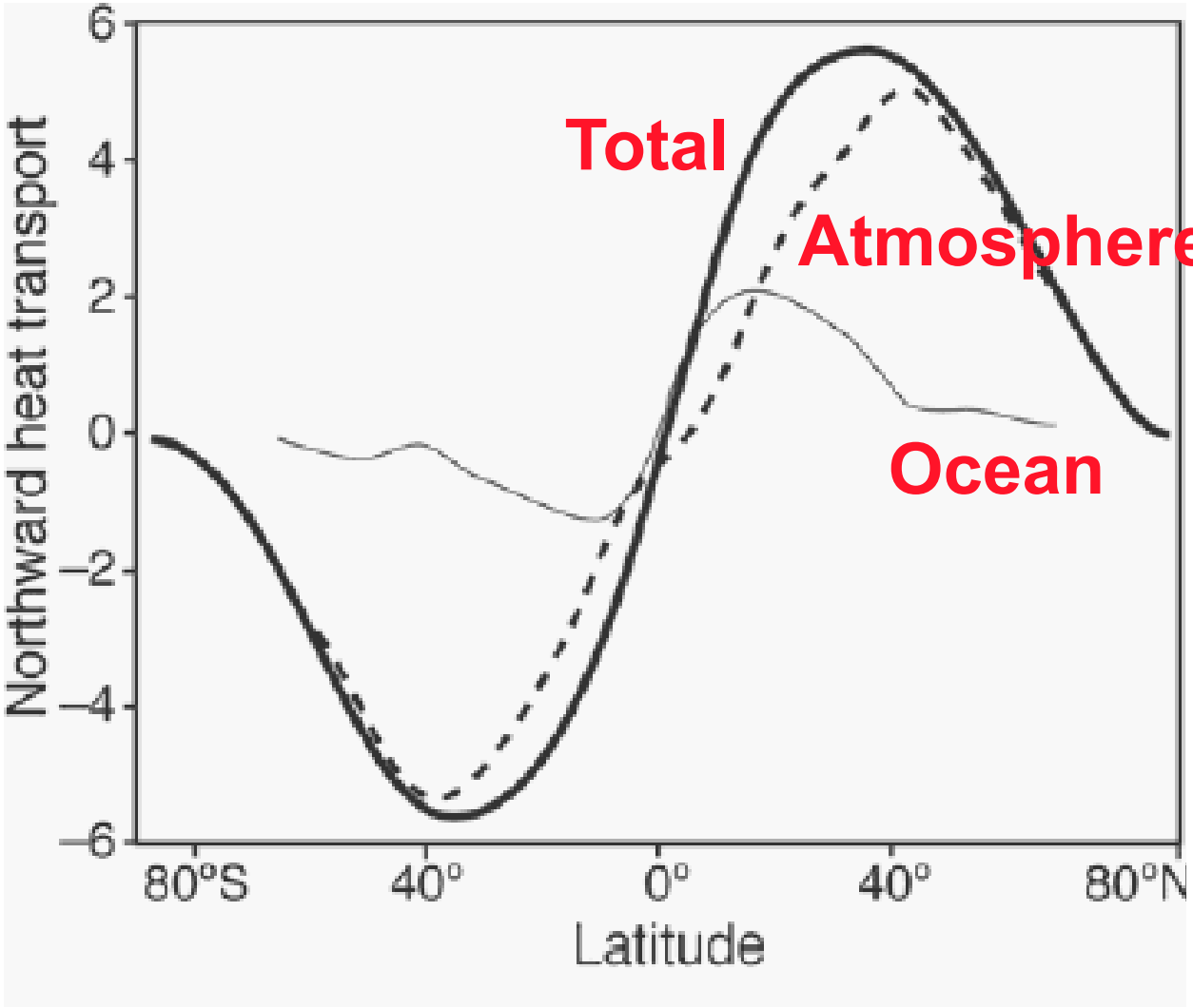
Energy Transport



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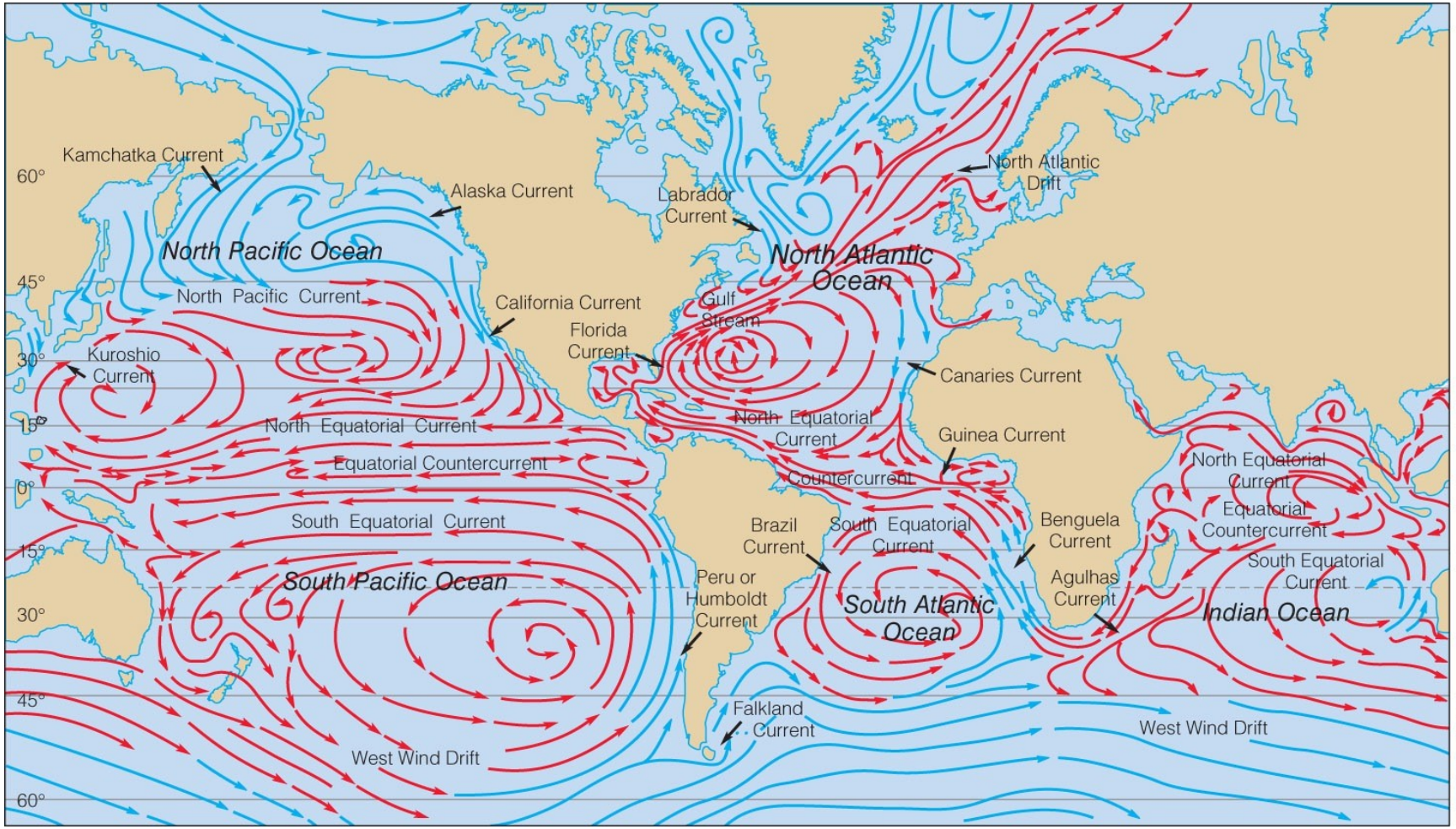
At present, Northern Hemisphere poleward heat transport by oceans is comparable to but less than that of the atmosphere.

(Meridional) North-South Energy Transport



In the Southern Hemisphere, atmosphere does most of the heat transport.

Surface Water Circulation



Gyres are formed through winds and friction, Earth's rotation, with the location of land masses blocking the circulation

Instabilities



Thermal Inertia: Land surfaces heat up faster than water

Thermal Properties: Land vs. Ocean

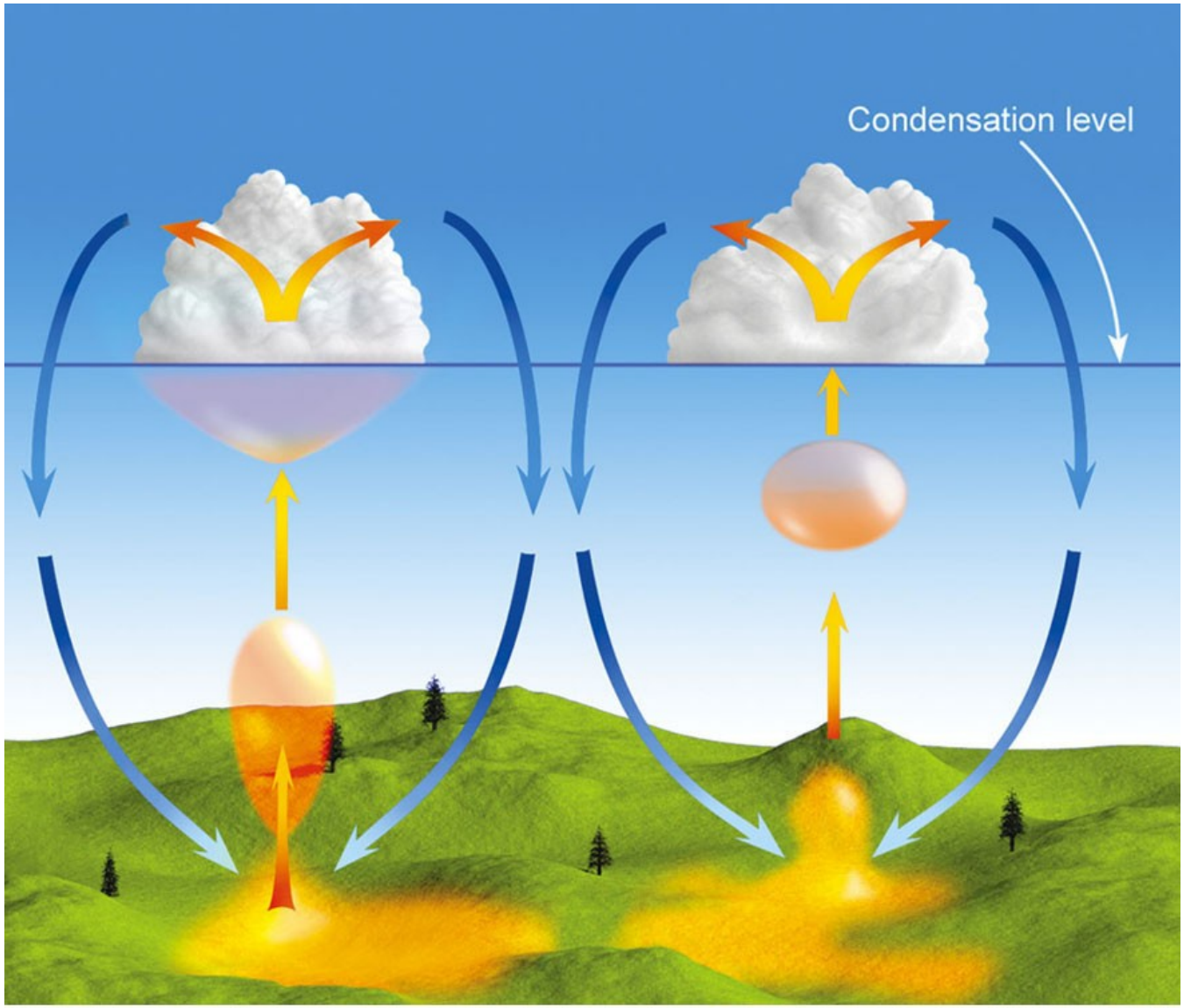
When sunlight heats the ground surface, only a thin layer gets warmed. Because of mixing, that layer is much thicker for oceans.

→ Land areas both warm and cool faster than the ocean surface;

→ Higher diurnal and seasonal temperature range.

→ “Continental” vs “maritime” climates.

Surface Heating; Cooling Aloft

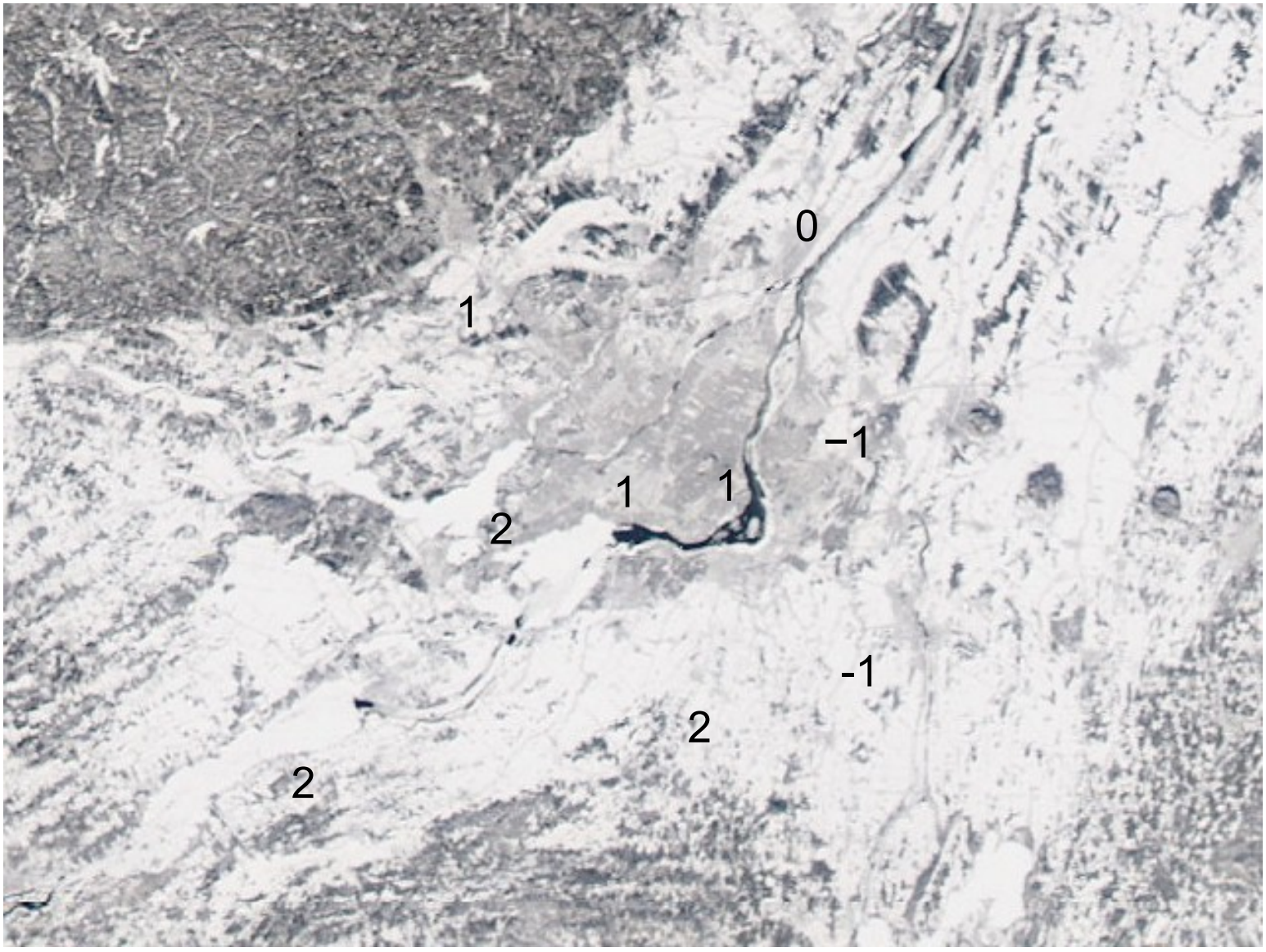


© 2007 Thomson Higher Education

Warm land heats the air above it

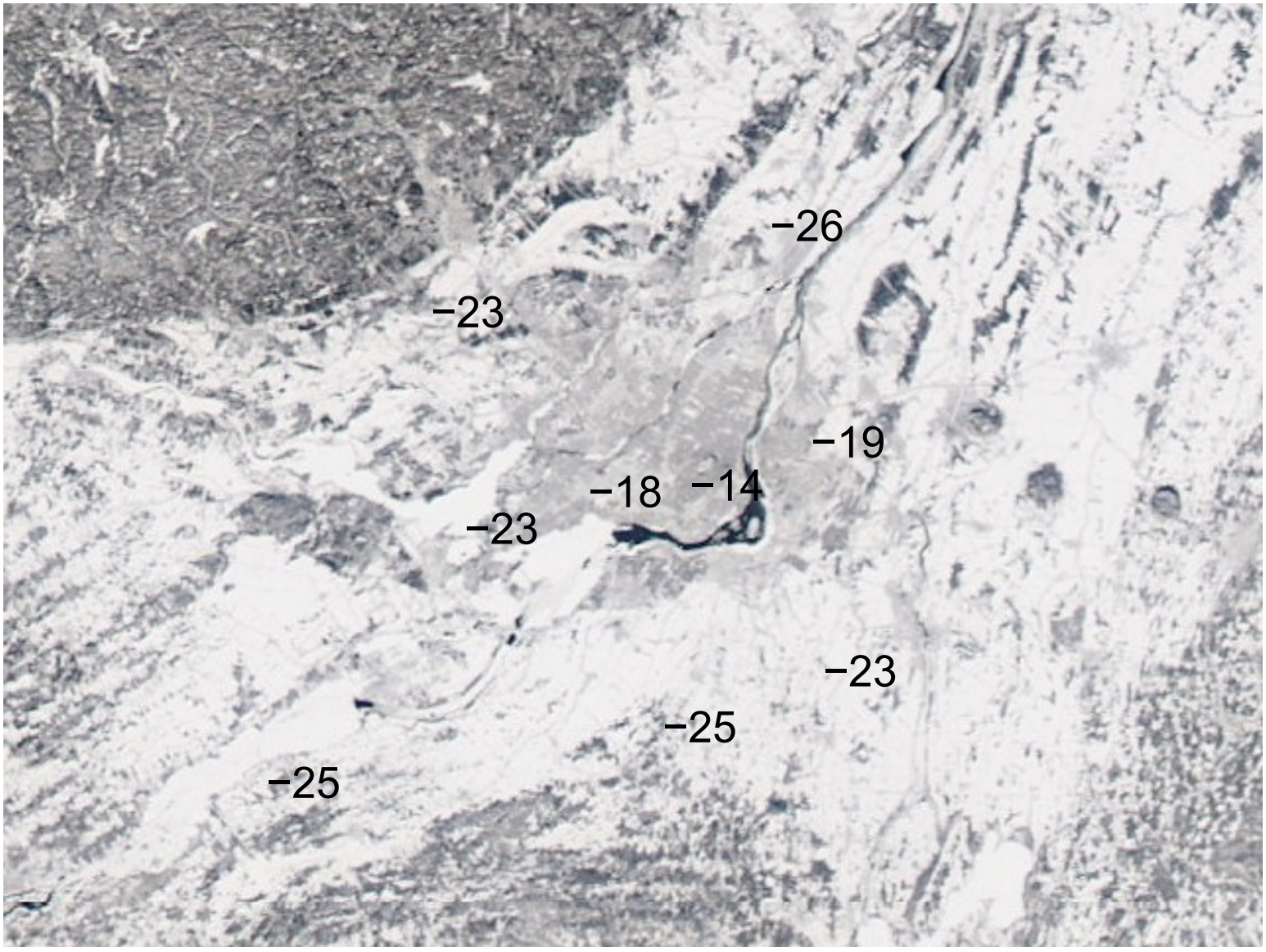
Ex.: Urban Heat Islands

Temperature at midnight last Decmber
No clear signal because of good winds, clouds

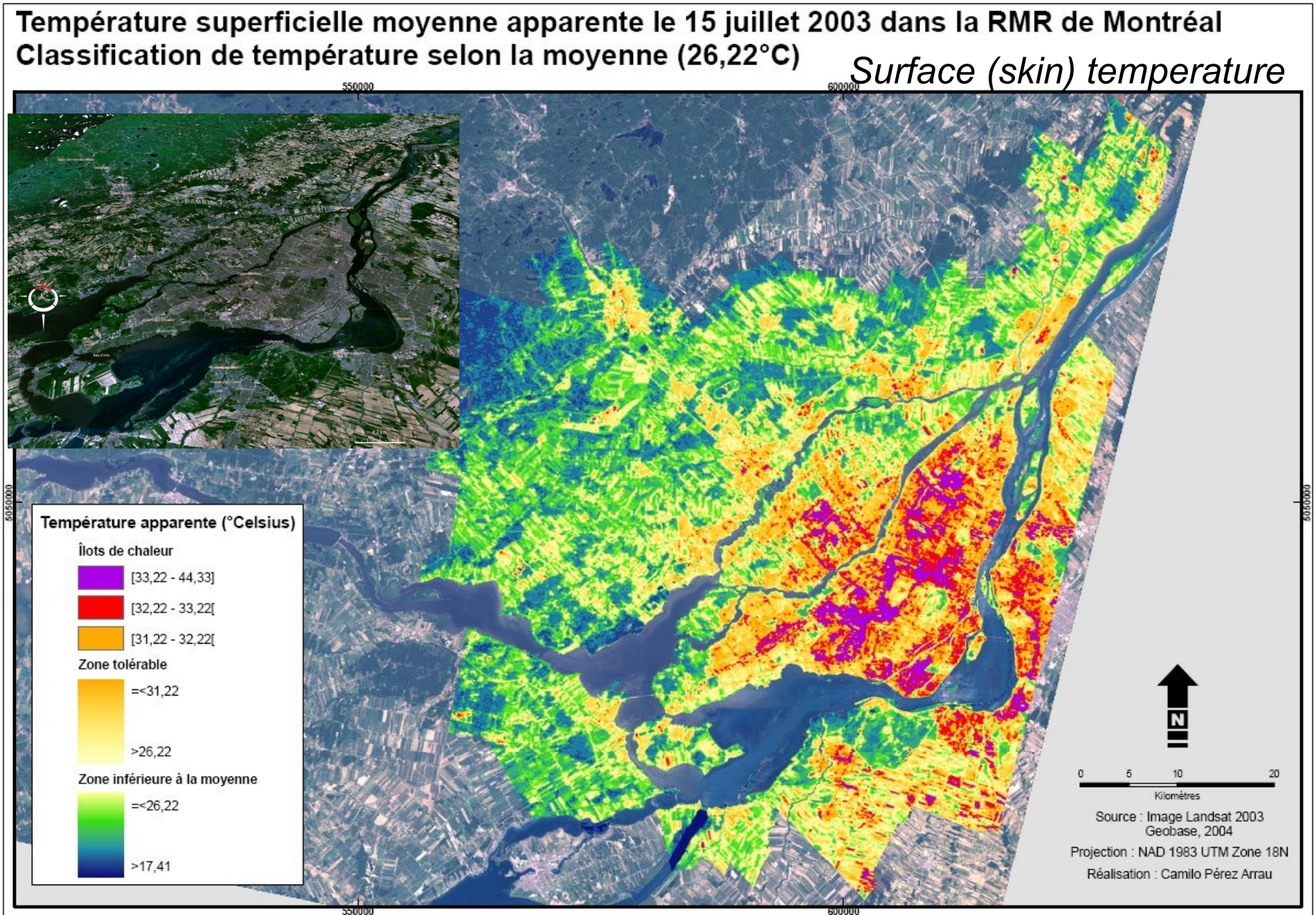


Ex.: Urban Heat Islands

Minimum temperatures for last year in January



Ex.: Urban Heat Islands



Other Land / Ocean Effects

The influence of oceans on nearby land will also depend on the temperature of the water, warmer waters inducing milder, rainier climates.

In parallel, snow/ice stays better on land than on water. Reflectance/whiteness of snow furthers cooling in winter on continents.

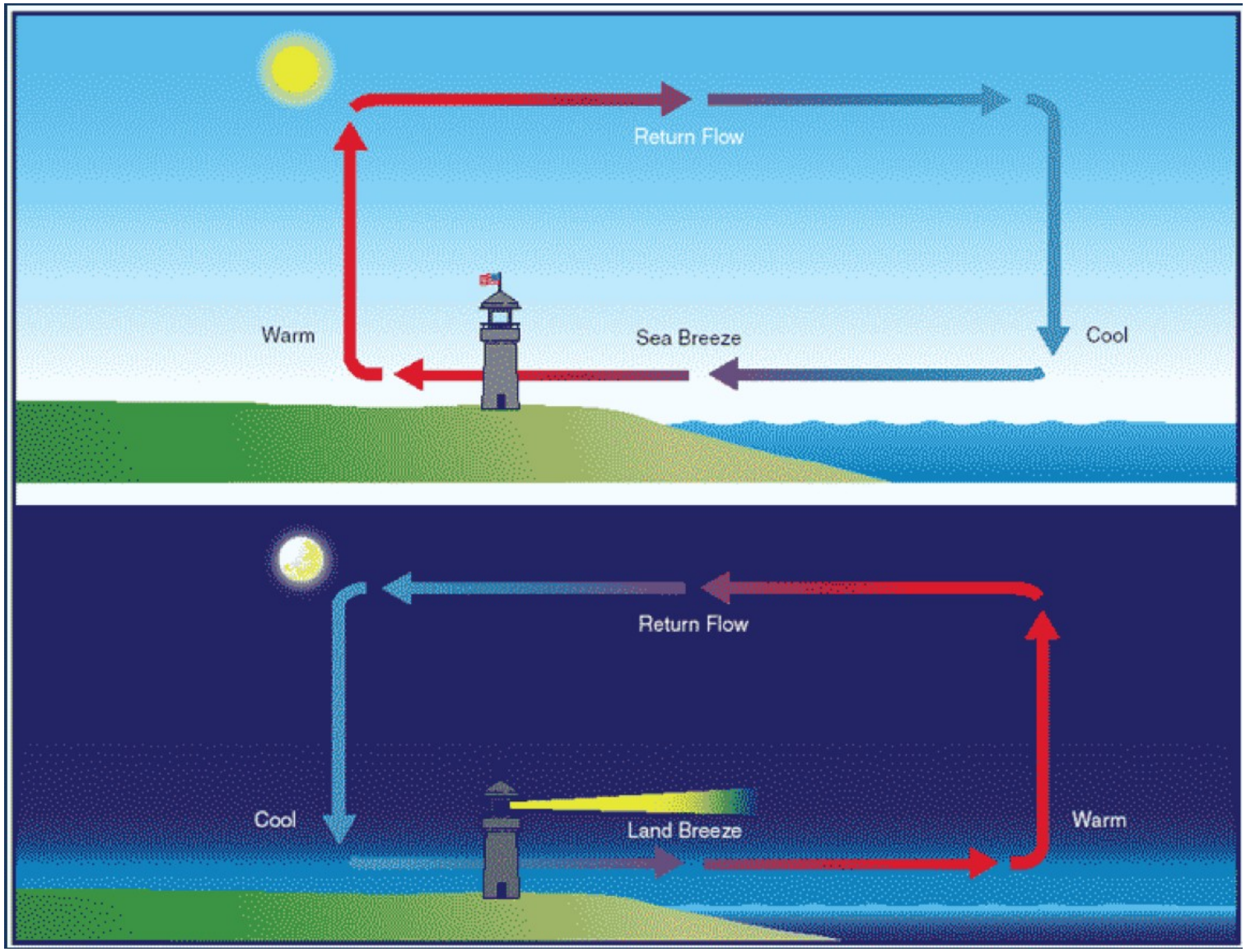
This is called Albedo


Sea & land breezes
Air over the sand is heated and rises

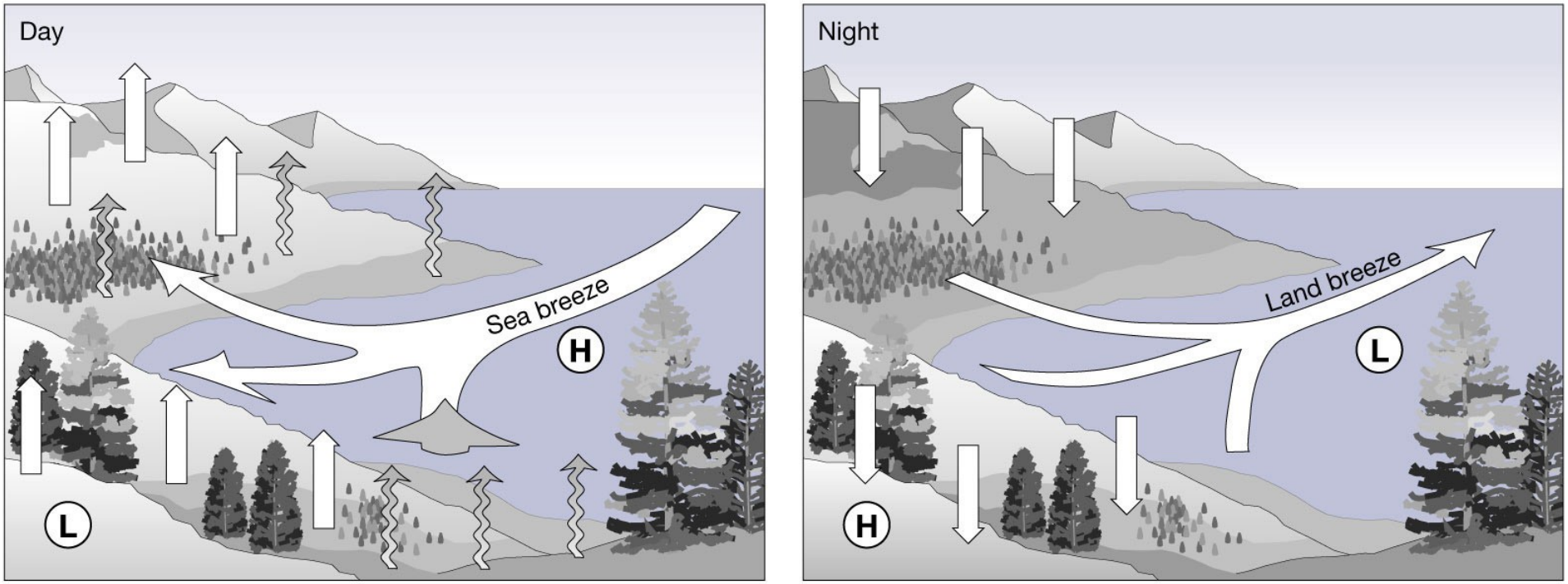
Sea & land breezes



Land and Sea Breezes

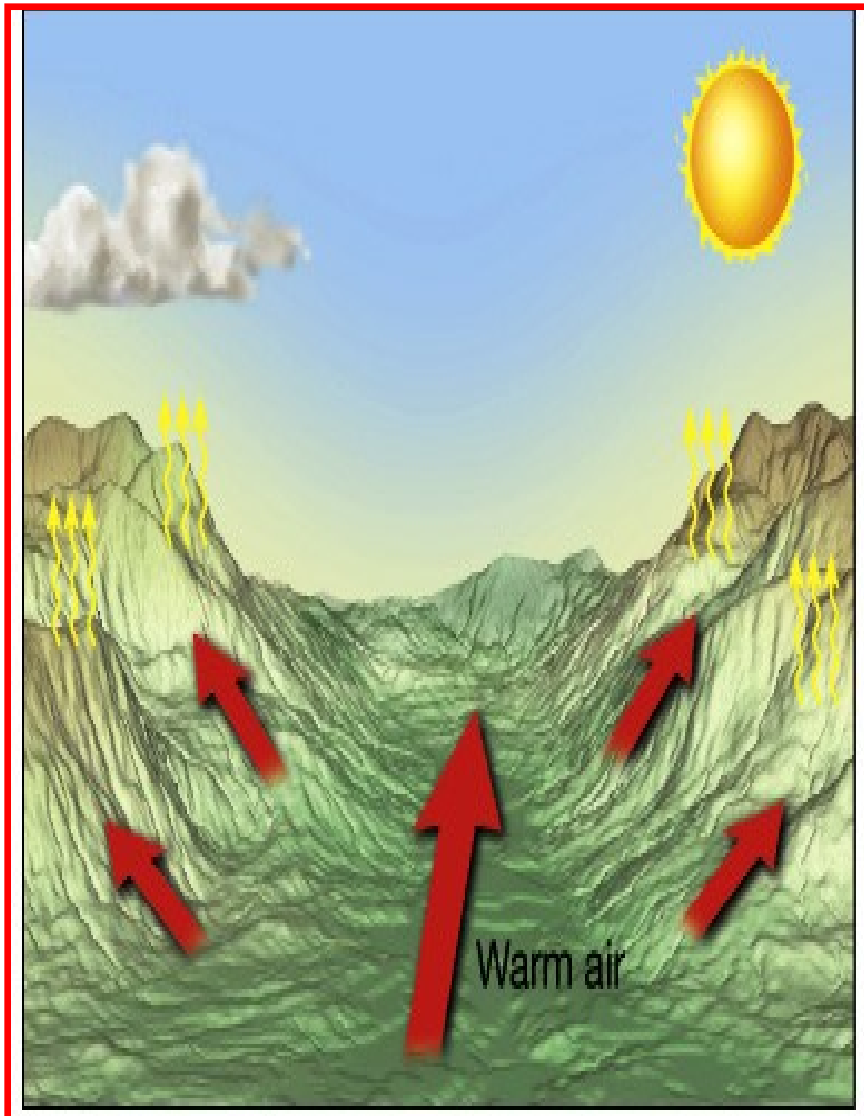


Temperature-Driven Circulations

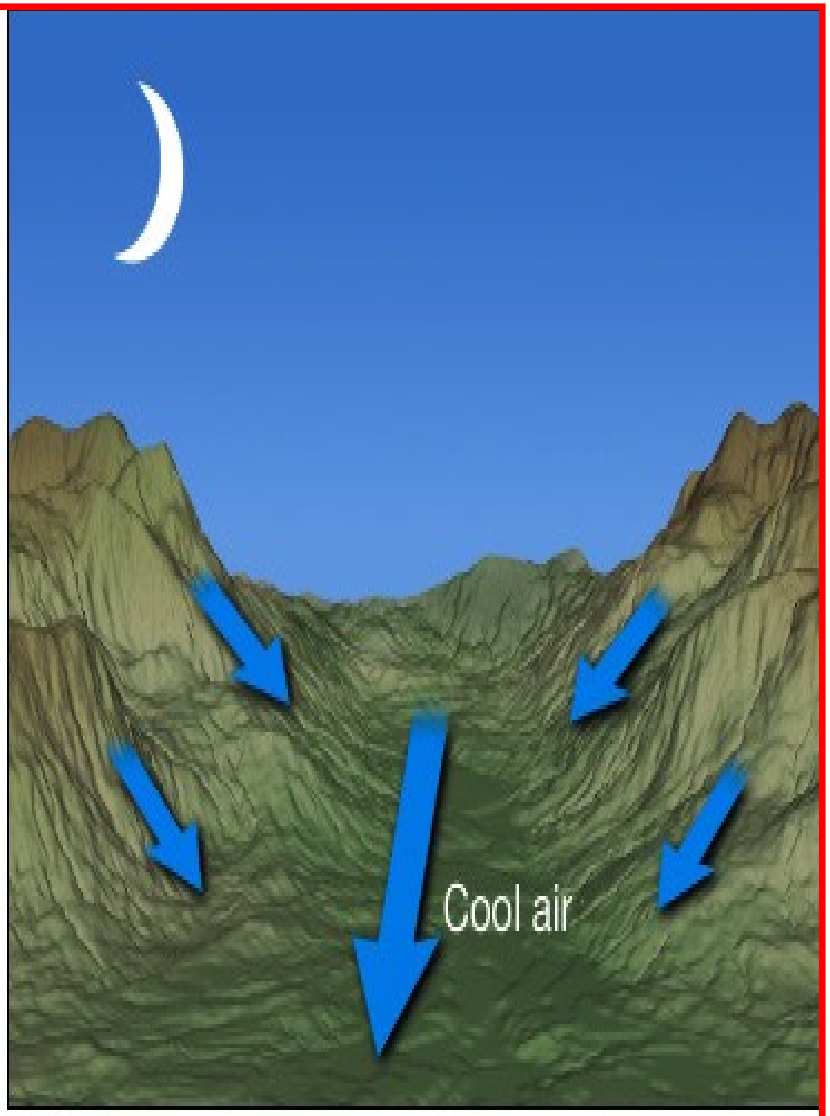


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When warm and cold regions are nearby, a thermally-driven circulation sets up, with a surface wind blowing from cold to warm areas.

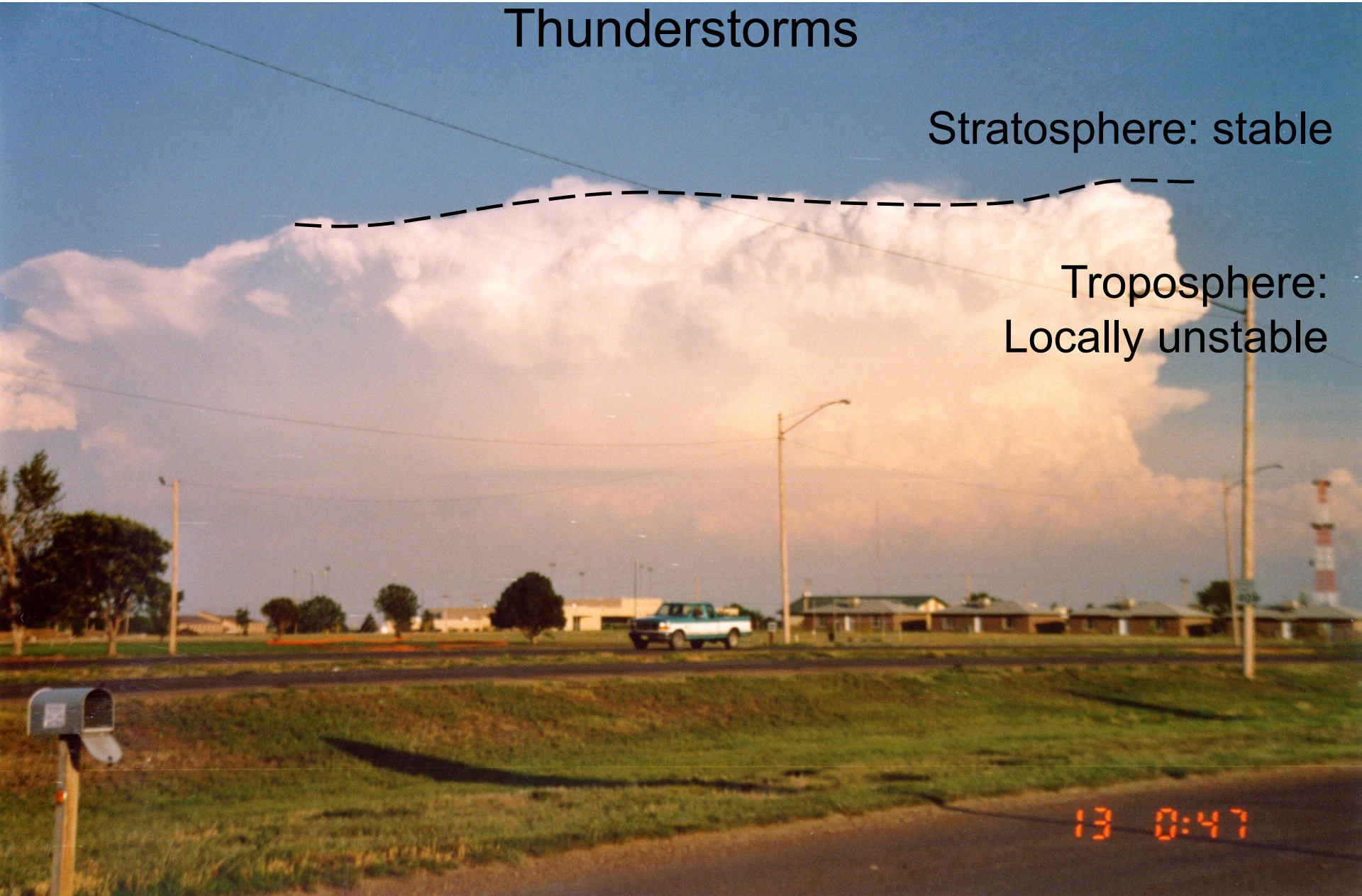


(a) Valley breeze



(b) Mountain breeze

Manifestation of this Instability: Thunderstorms

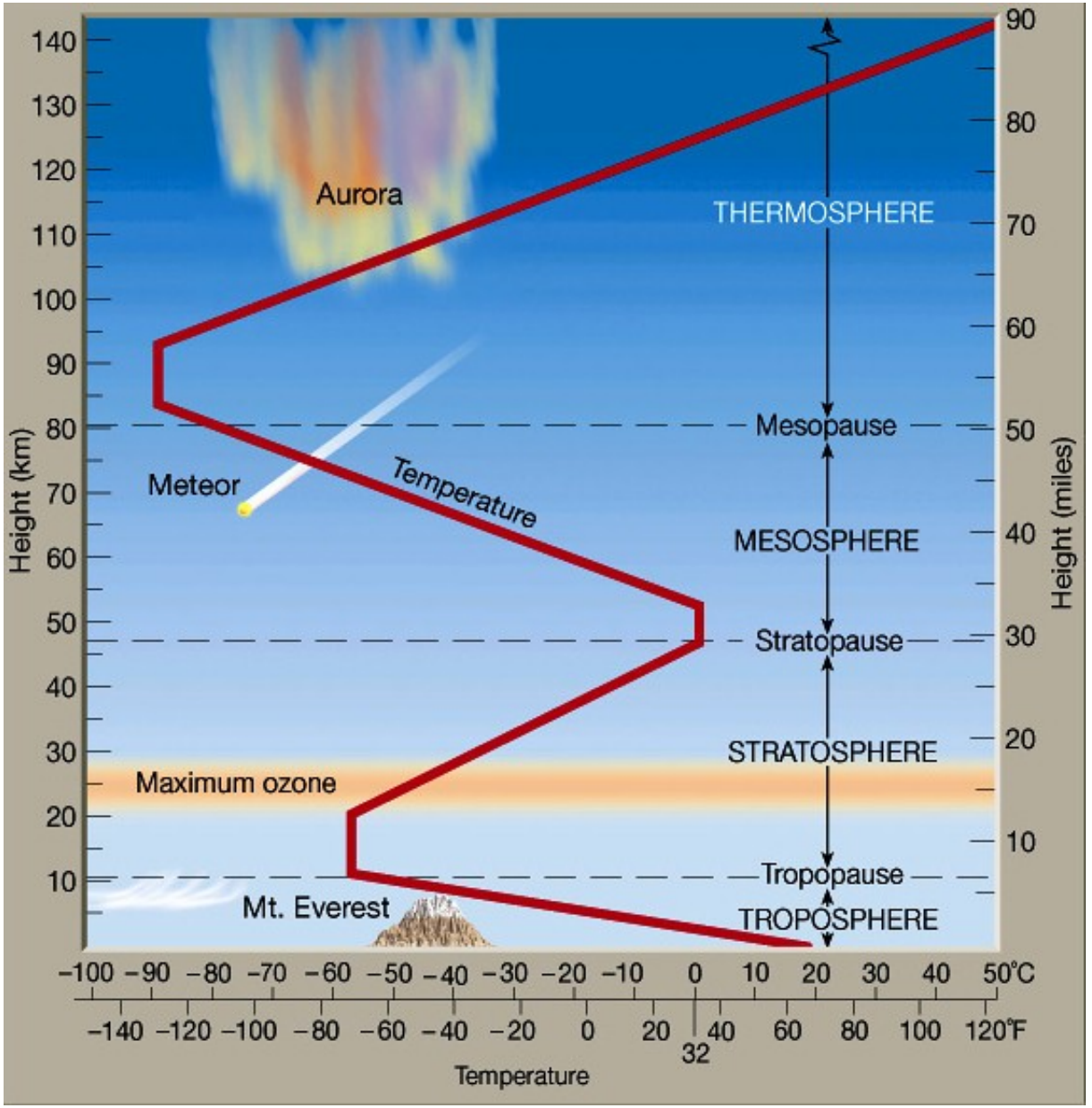


Layered Structure of the Atmosphere

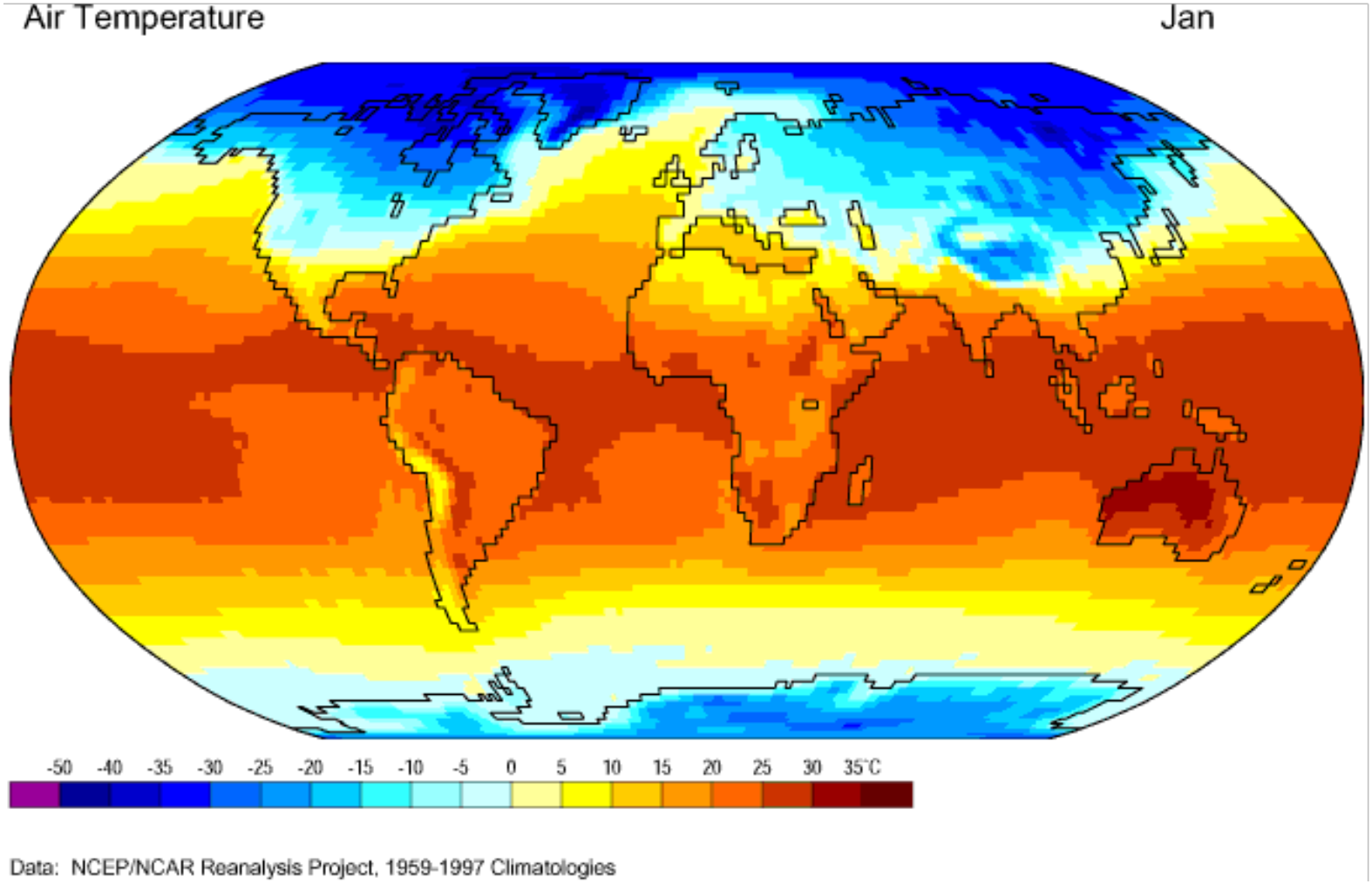
Layers are determined by how temperature changes with altitude.

The troposphere is where weather occurs (0 → 10-15 km);

Above lie the stratosphere, mesosphere and thermosphere.

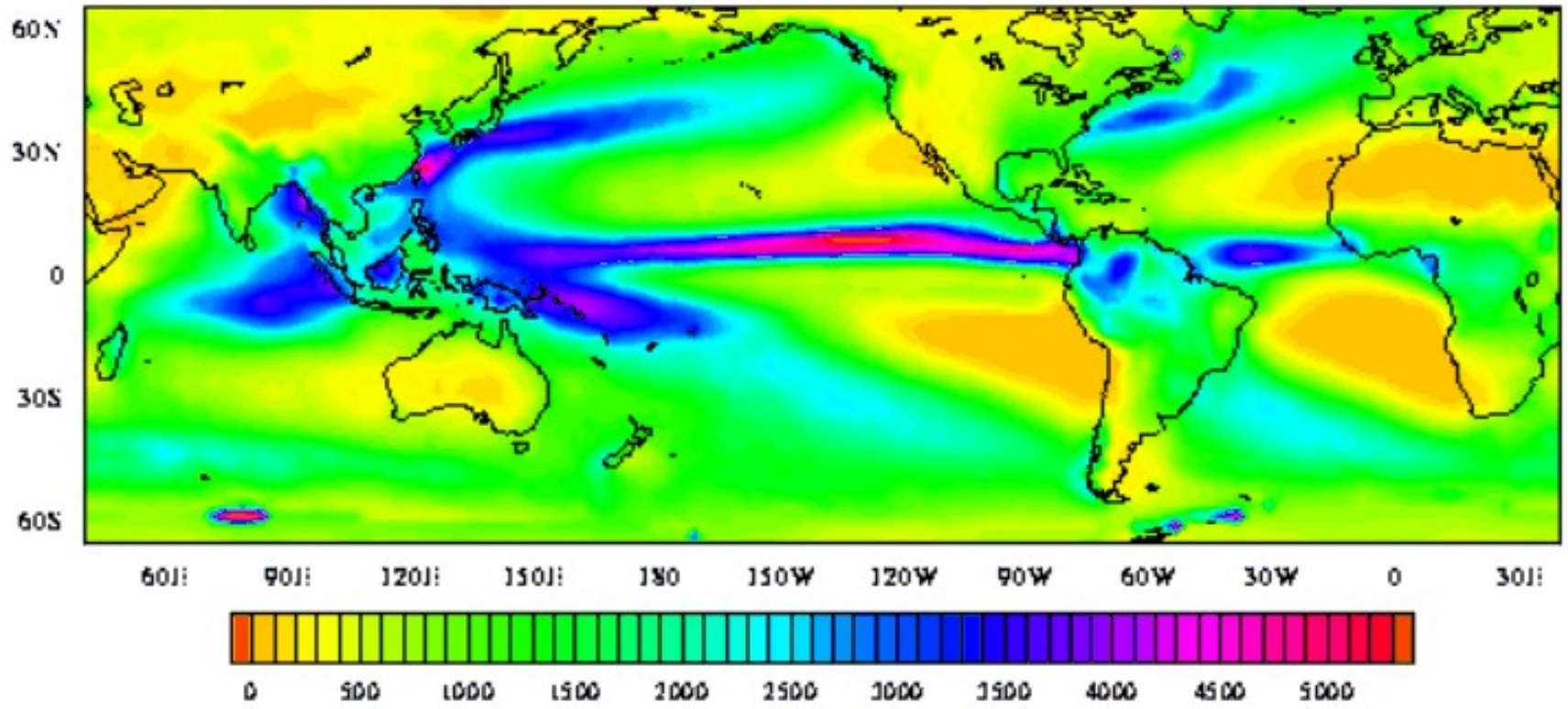


Surface Temperature

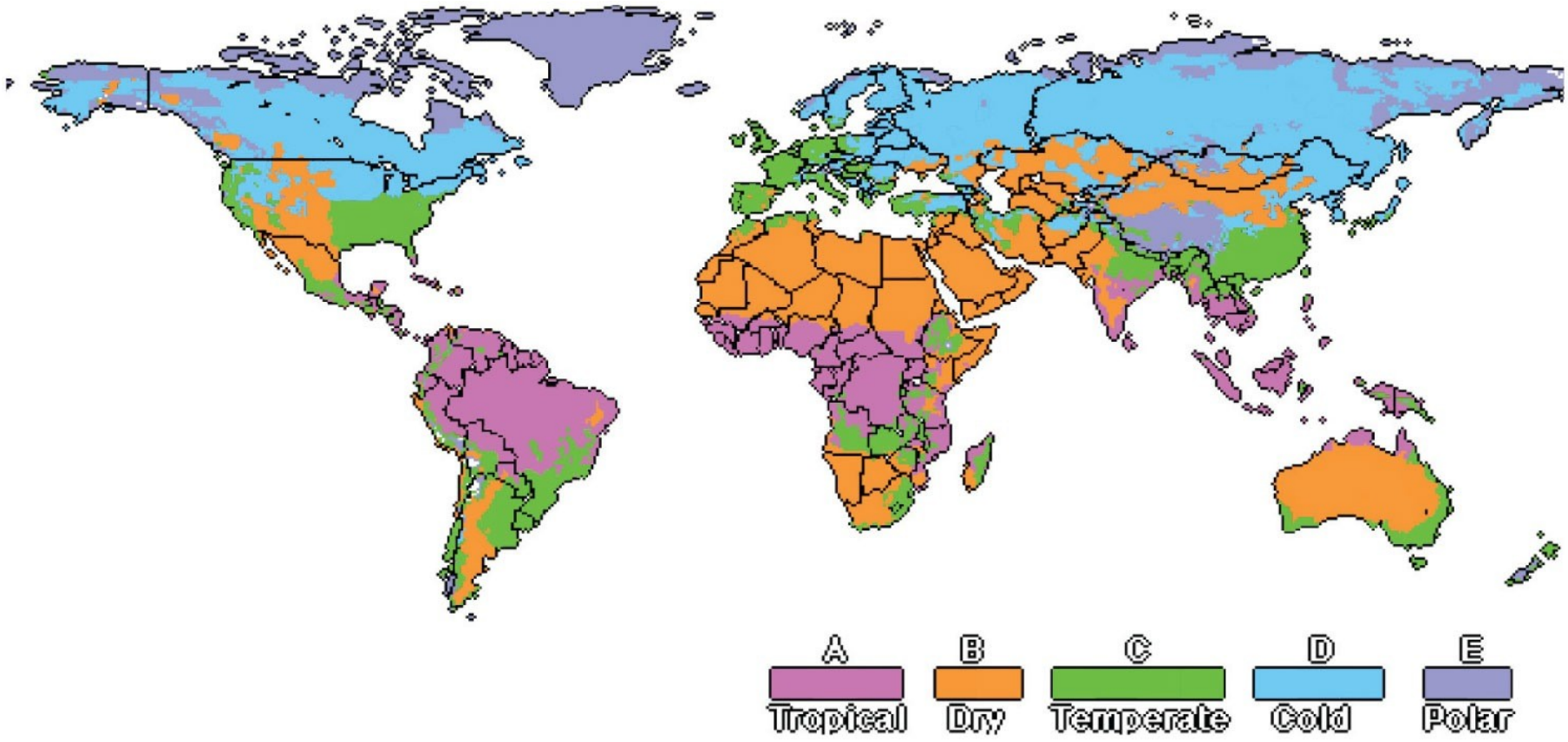


Net result. Living beings at a given location must adapt to the local peculiarities in climate.

Precipitation

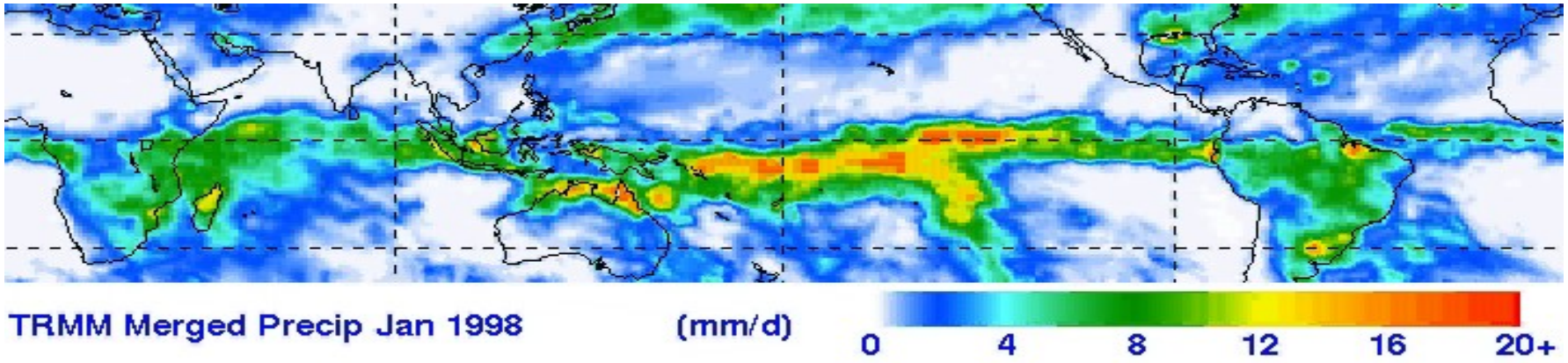


Note local maxima (minima) over warm (cold) currents and dry regions downwind of mountains; superposed on latitudinal pattern.



(From Kotteck, M., J. Griieser, C. Beck, B. Rudolf, and R. Rubel. 2006. World map of the Koppen classification updated. *Meteoroligische Zeitschrift* 15: 259–263.)

Precipitation ITCZ



January: Maximum precipitation is south of the Equator

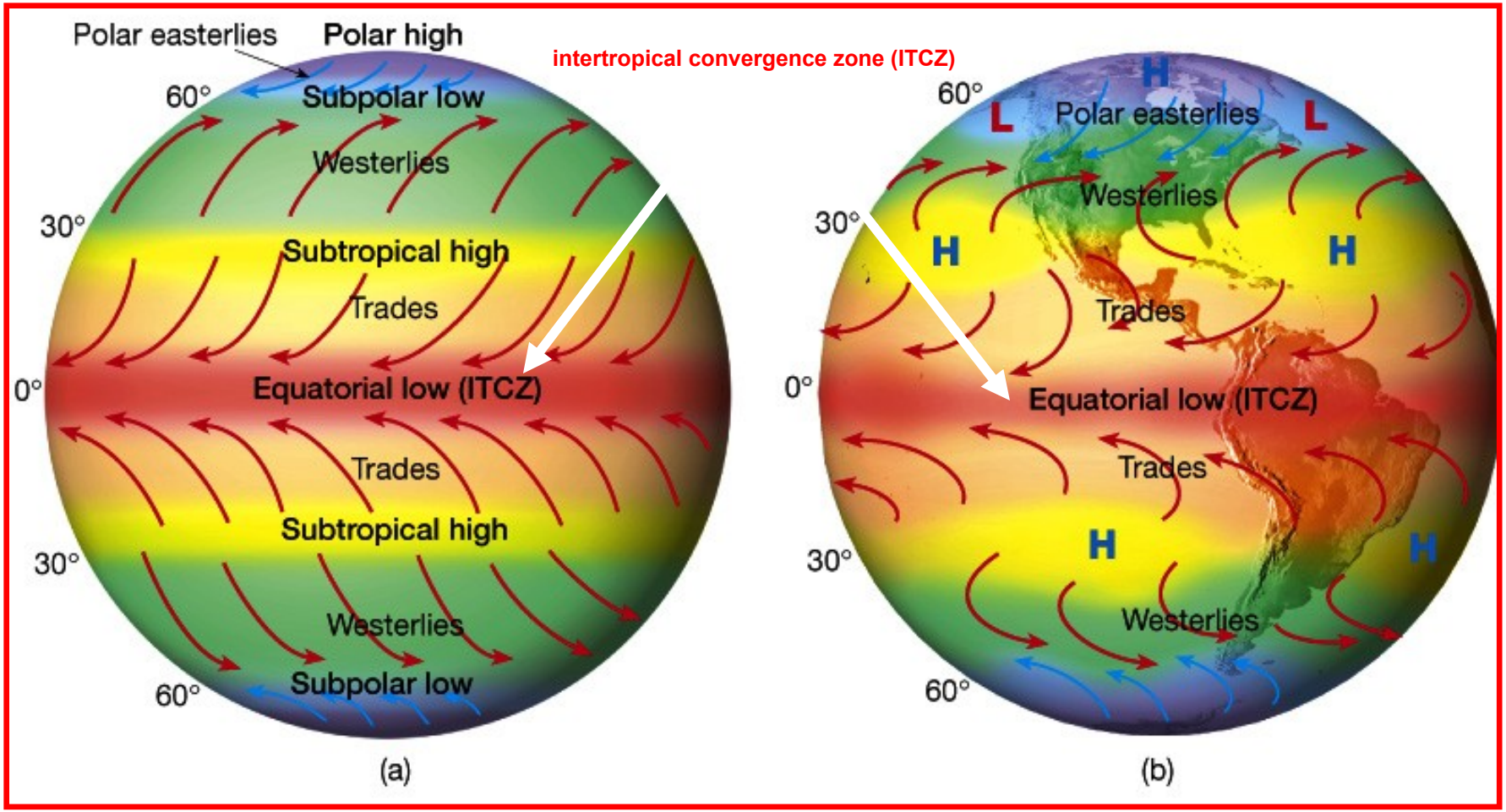

A
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Belize

Practice

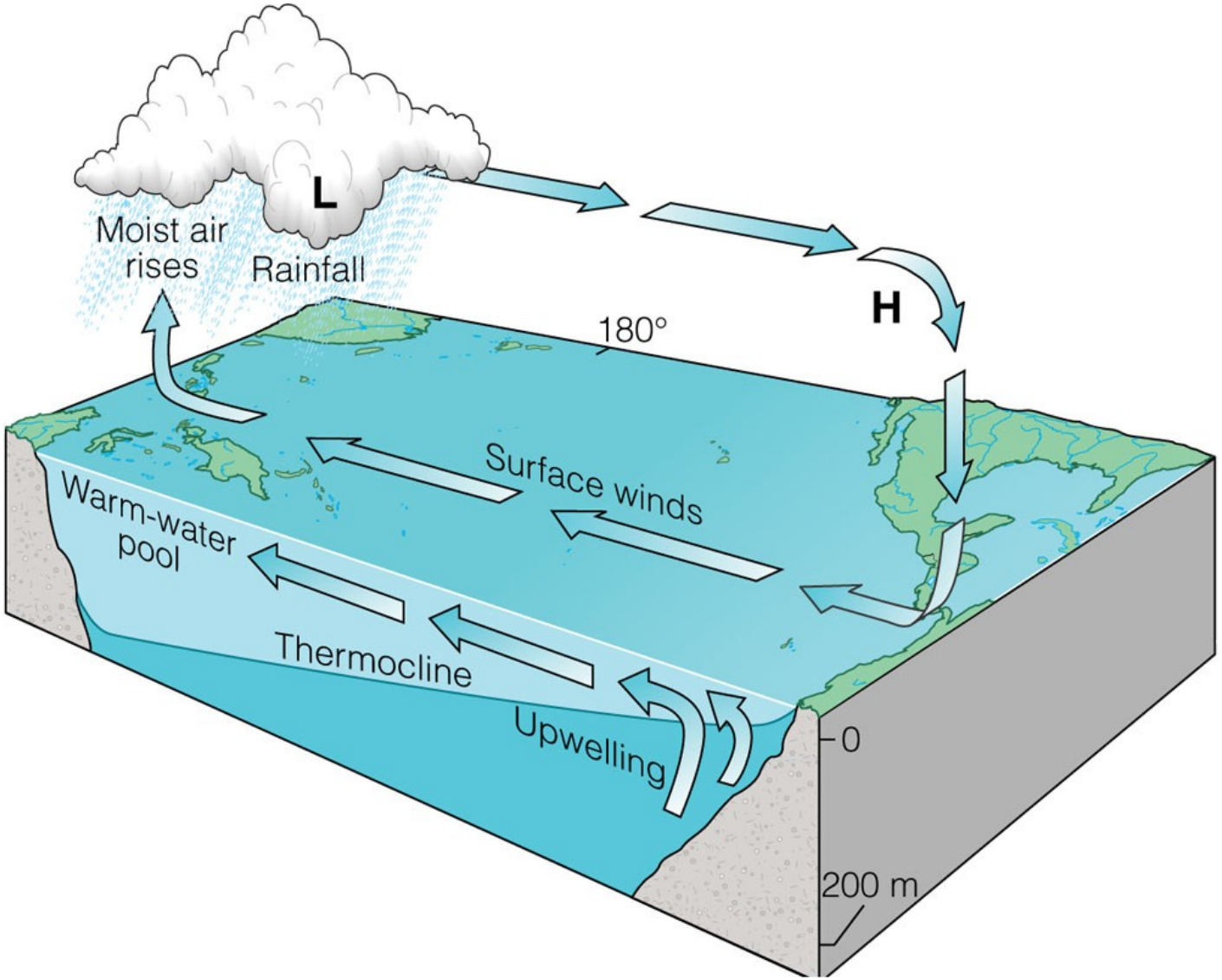
Pick a country that starts with the same letter as your first name:

What does the three-cell (Hadley cell) model predict its climate should be?

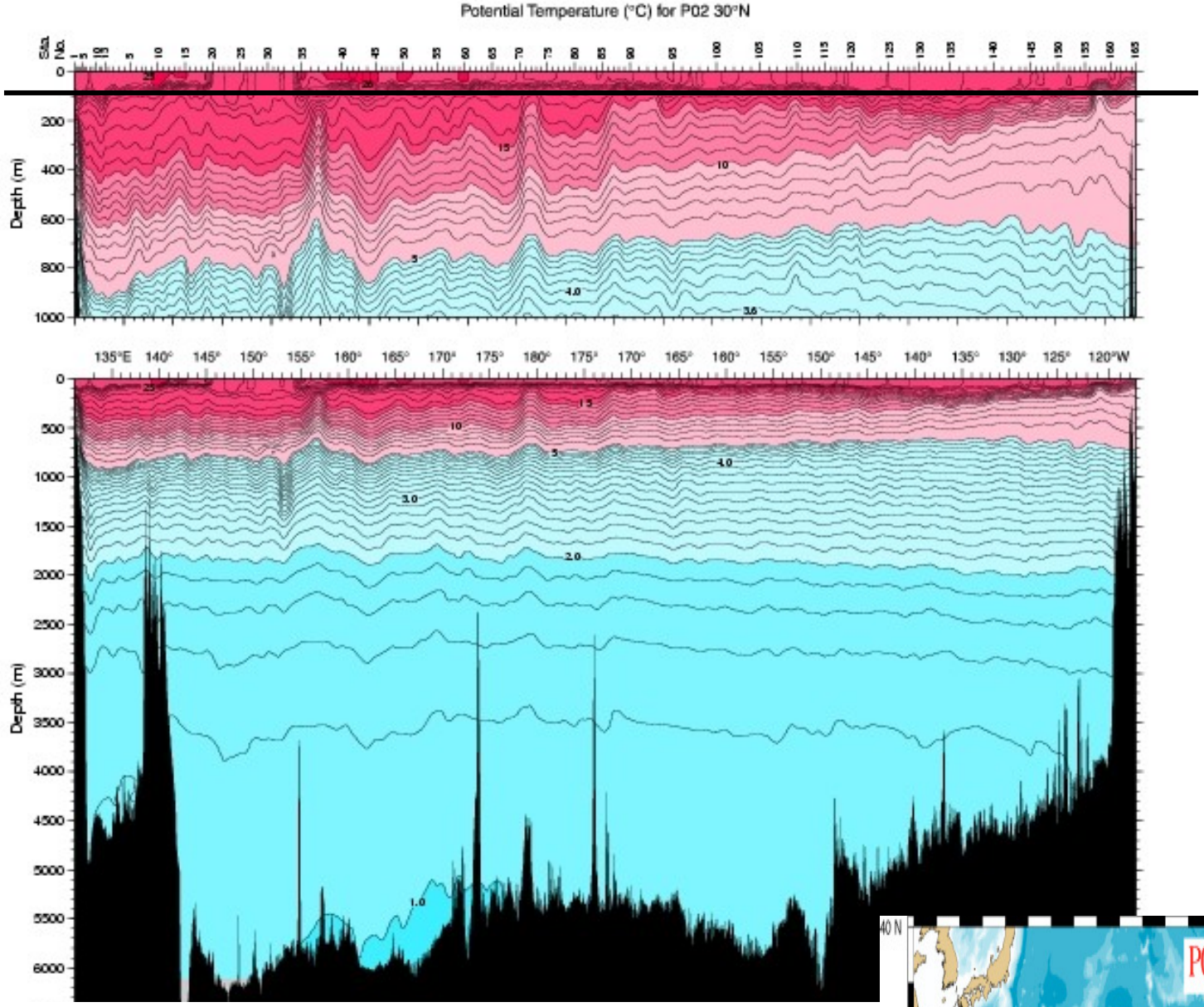
Will this change with the seasons?


Polar easterlies
Polar high
intertropical convergence zone (ITCZ)
60°
Subpolar low
Westerlies
30°
Subtropical high
Trades
0°
Equatorial low (ITCZ)
Trades
Subtropical high
30°
Westerlies
60°
Subpolar low
(a)
H
60°
L
Polar easterlies
L
Westerlies
30°
H
H
Trades
0°
Equatorial low (ITCZ)
Trades
H
H
30°
Westerlies
60°
(b)

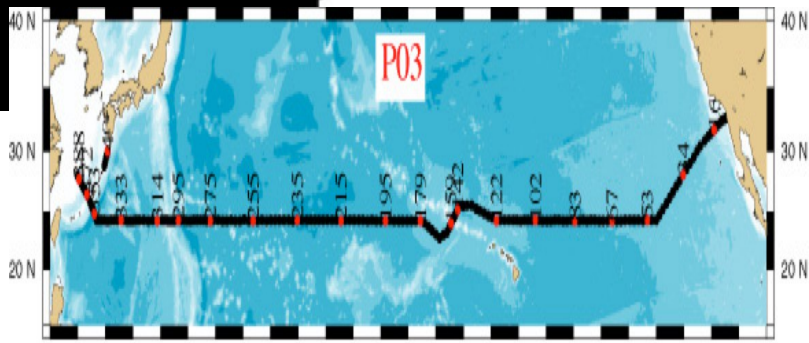
Surface winds Drive Ocean Currents



© 2005 Brooks/Cole - Thomson



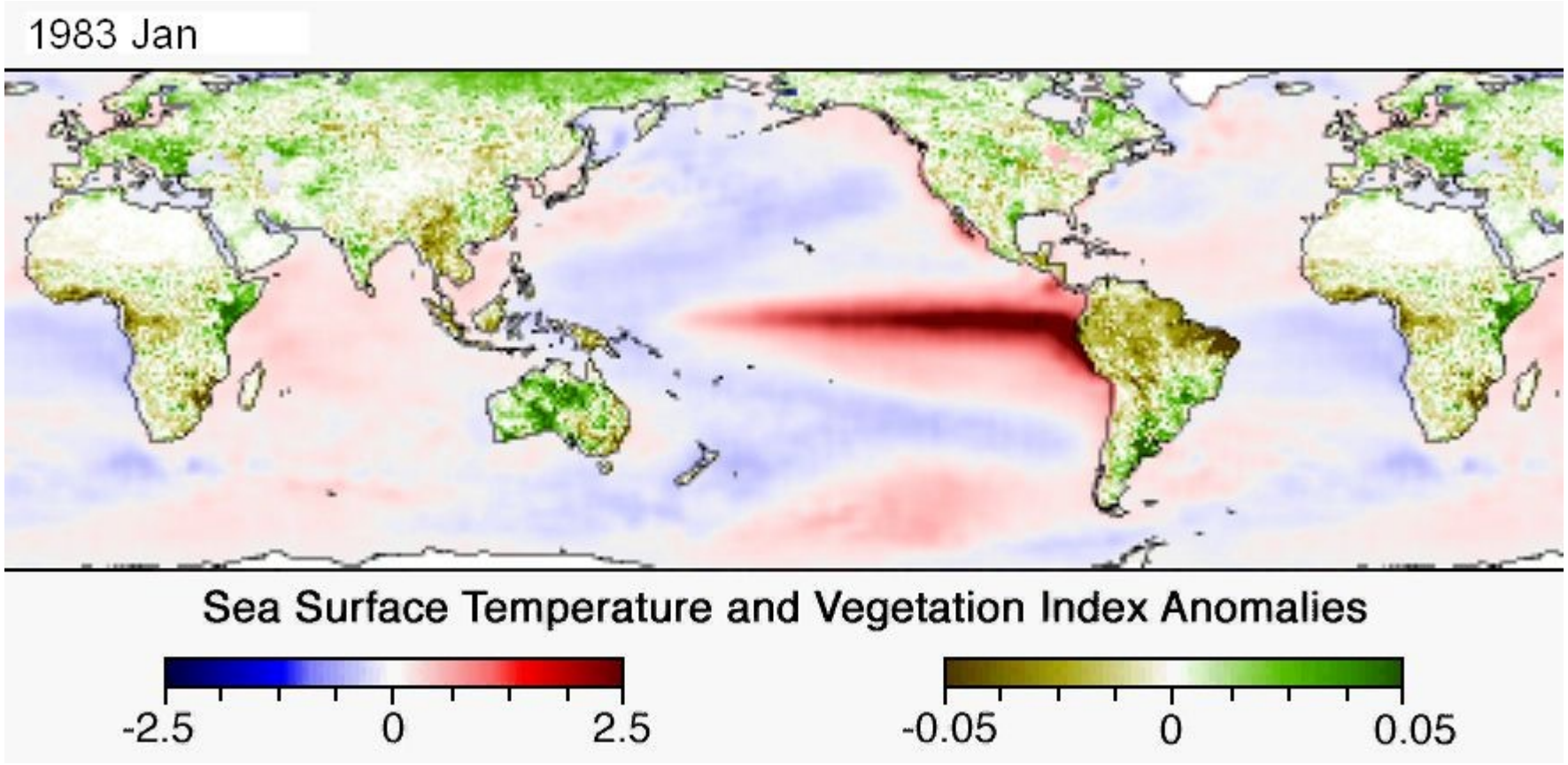
Ocean Temperature across the Pacific



From WHP Pacific Atlas (Talley, 2007)

http://www-pord.ucsd.edu/whp_atlas

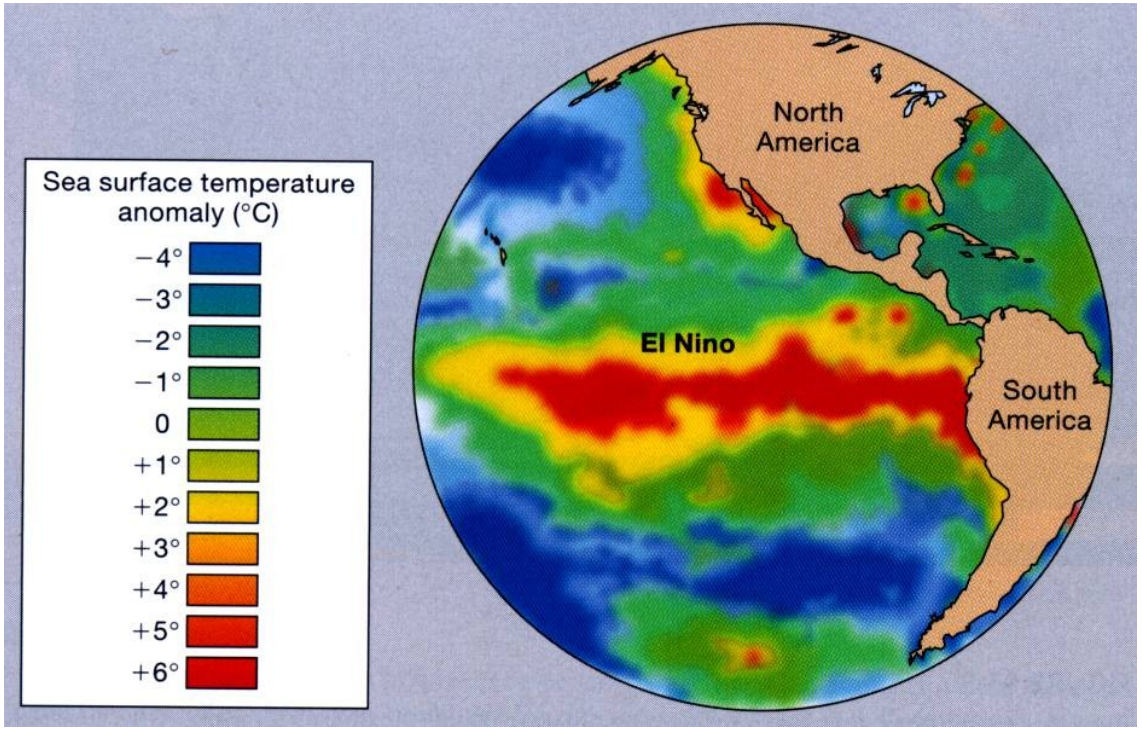
Oscillations in Air-Sea system



In addition to instabilities, *oscillations* occur in the atmosphere-ocean system.

The strongest is known as El Niño – Southern oscillation.

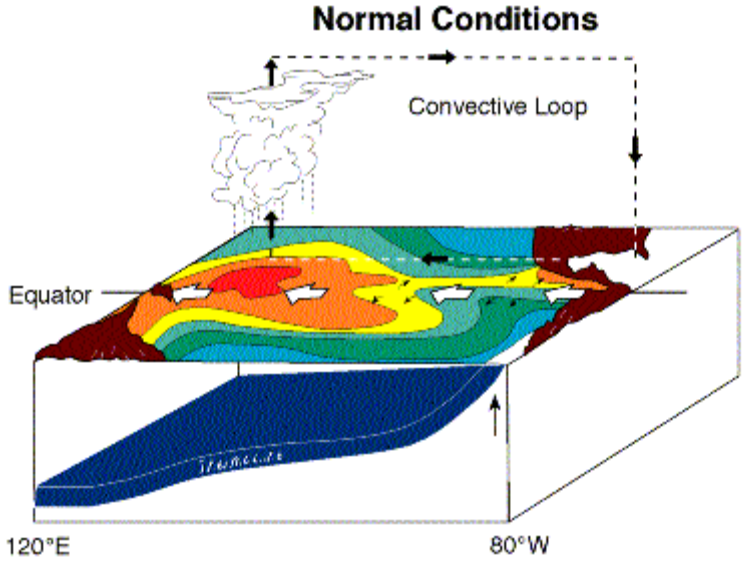
El Niño – Southern Oscillation (ENSO)



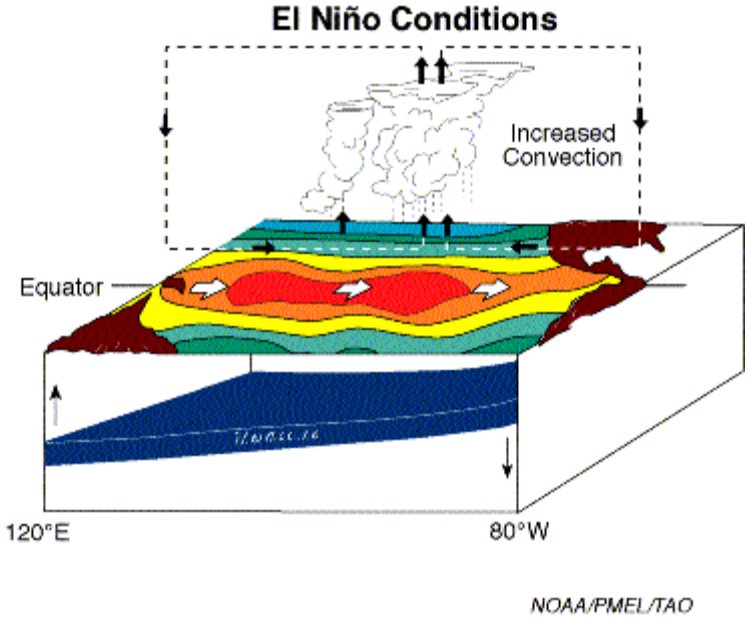
Occurs irregularly every 3-7 years, a result of complex air-sea interaction in the tropical Pacific;

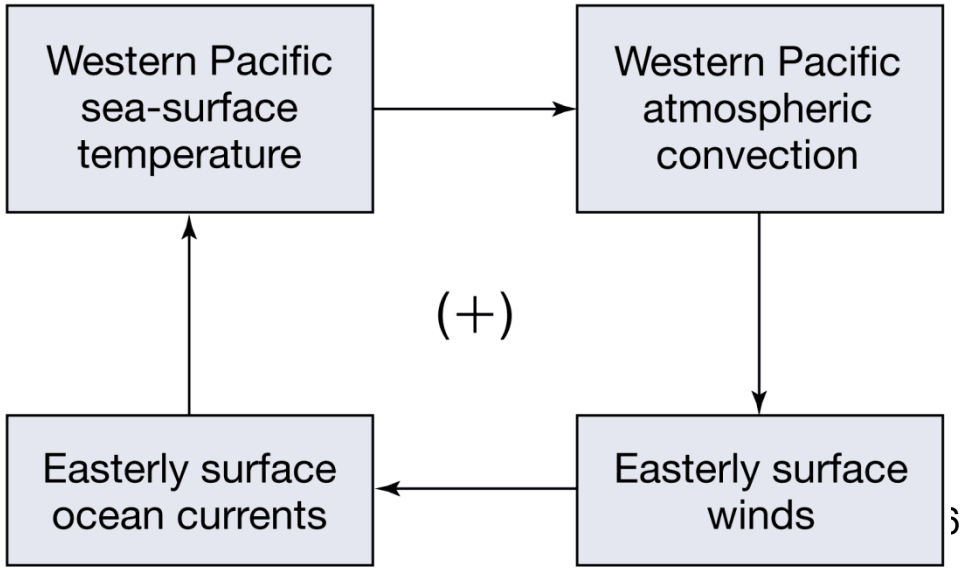
During El Niño, a warm pool of surface waters in the western tropical Pacific extends far to the east, to the Peruvian coast.

El Niño – Southern Oscillation (ENSO)

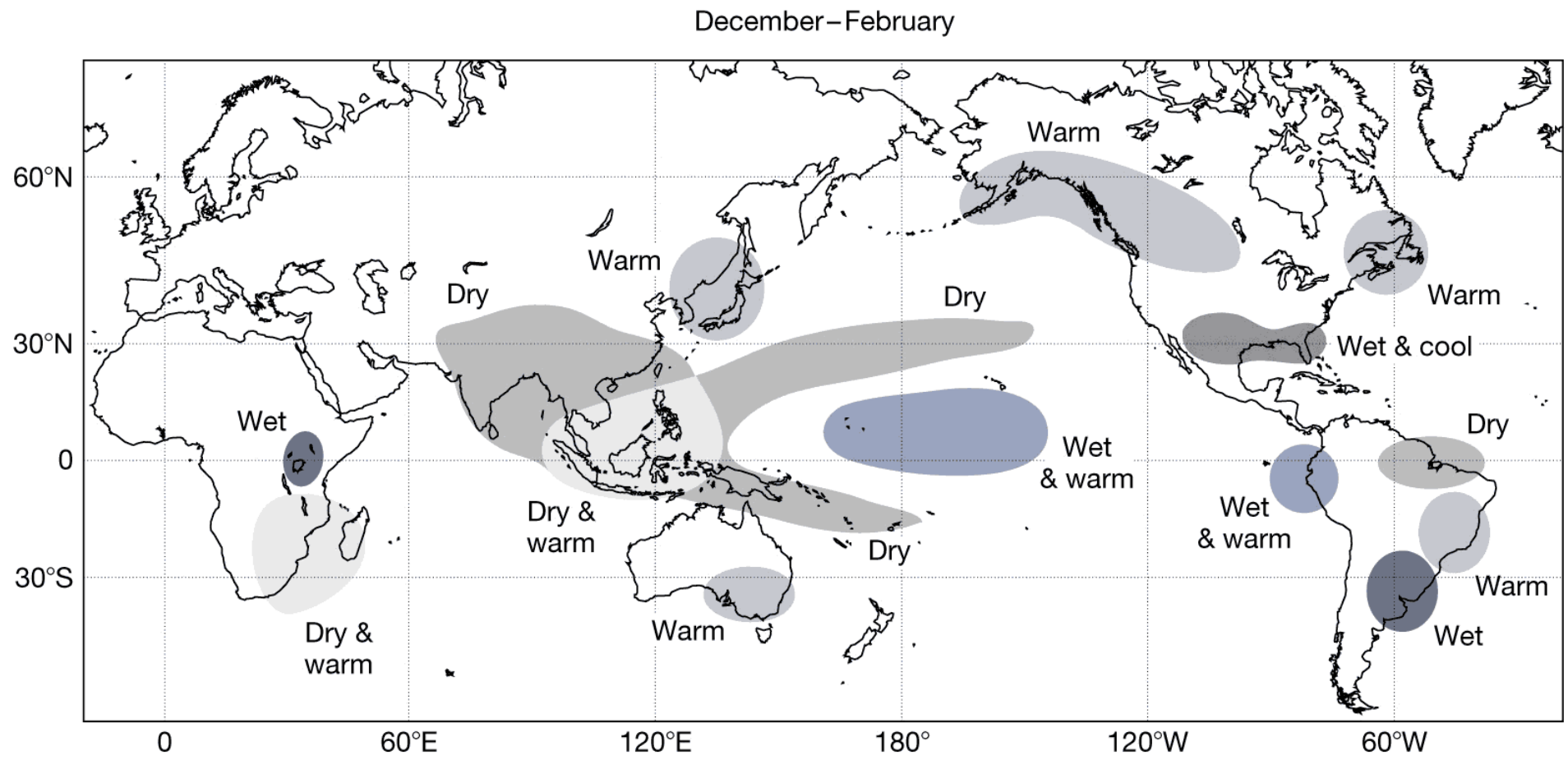


Warm waters in East Pac.
→ Displaces a normally rainy area from the west to central Pacific;
→ Reduces trade winds;
→ Feedback.



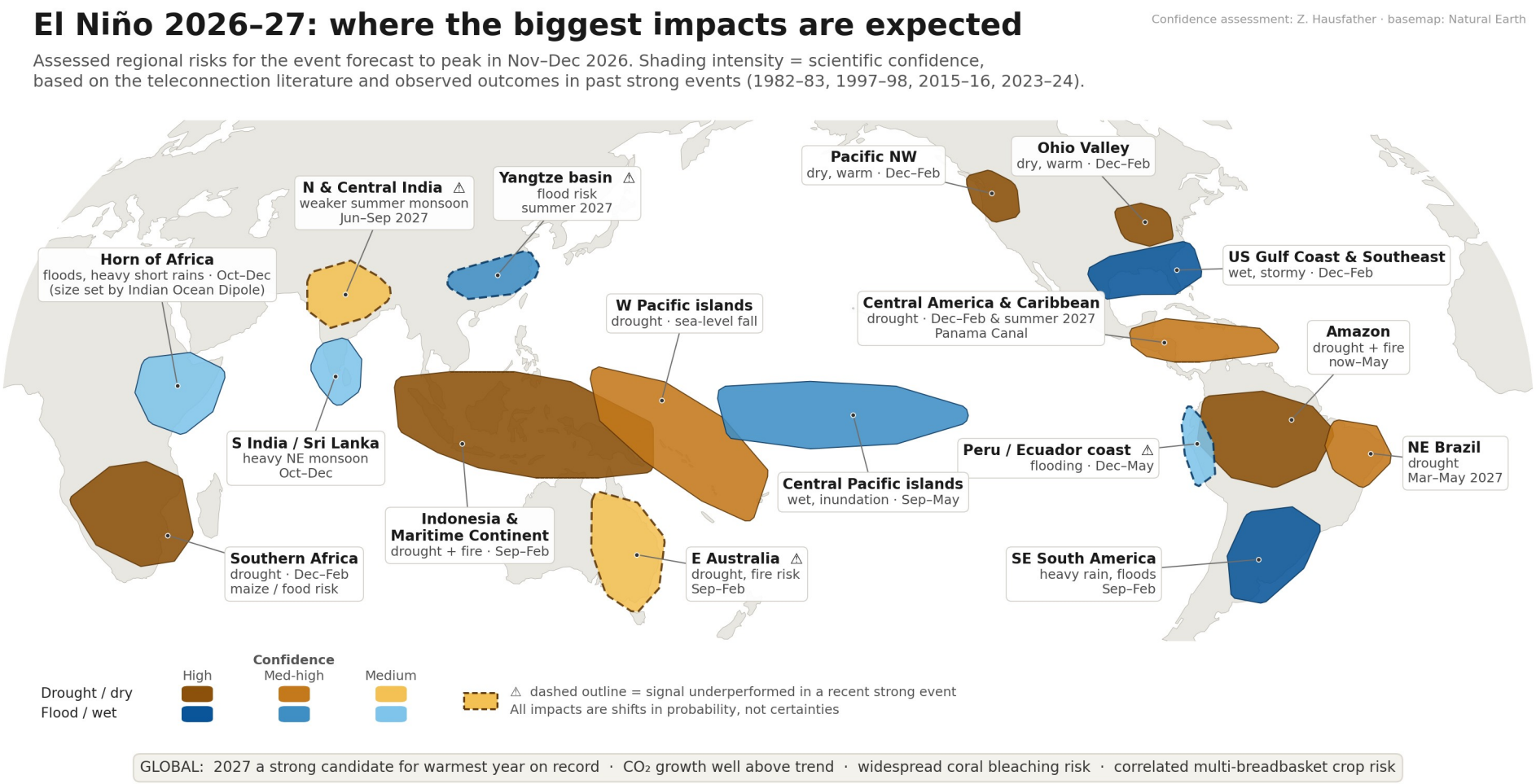


ENSO – Planetary-Wide Impacts



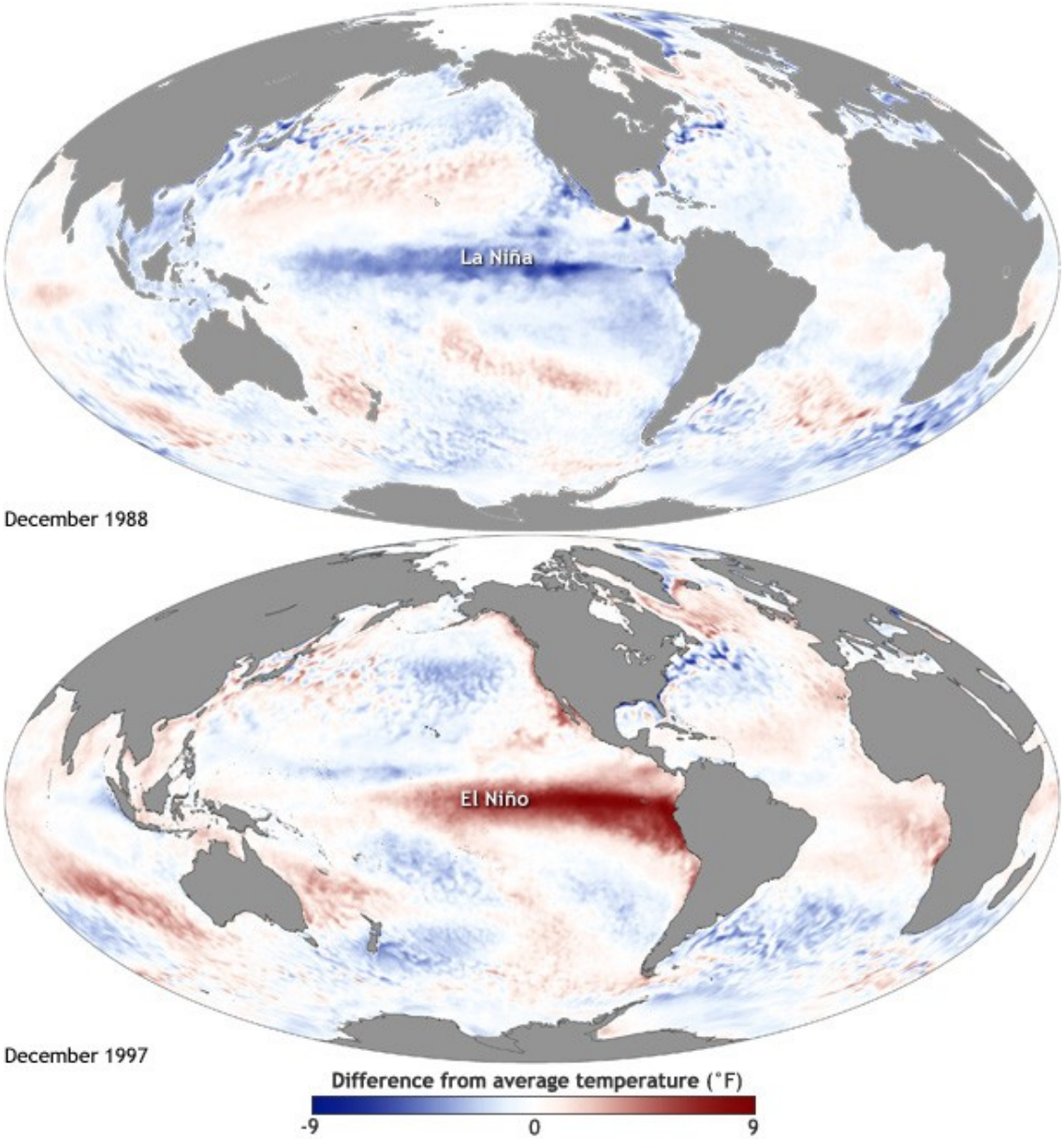
- Temp. and precip. anomalies in the tropics and elsewhere;
- 1982-83 El Niño: thousands of deaths (floods, droughts, disease, …) and over $13B damage.

ENSO – Planetary-Wide Impacts

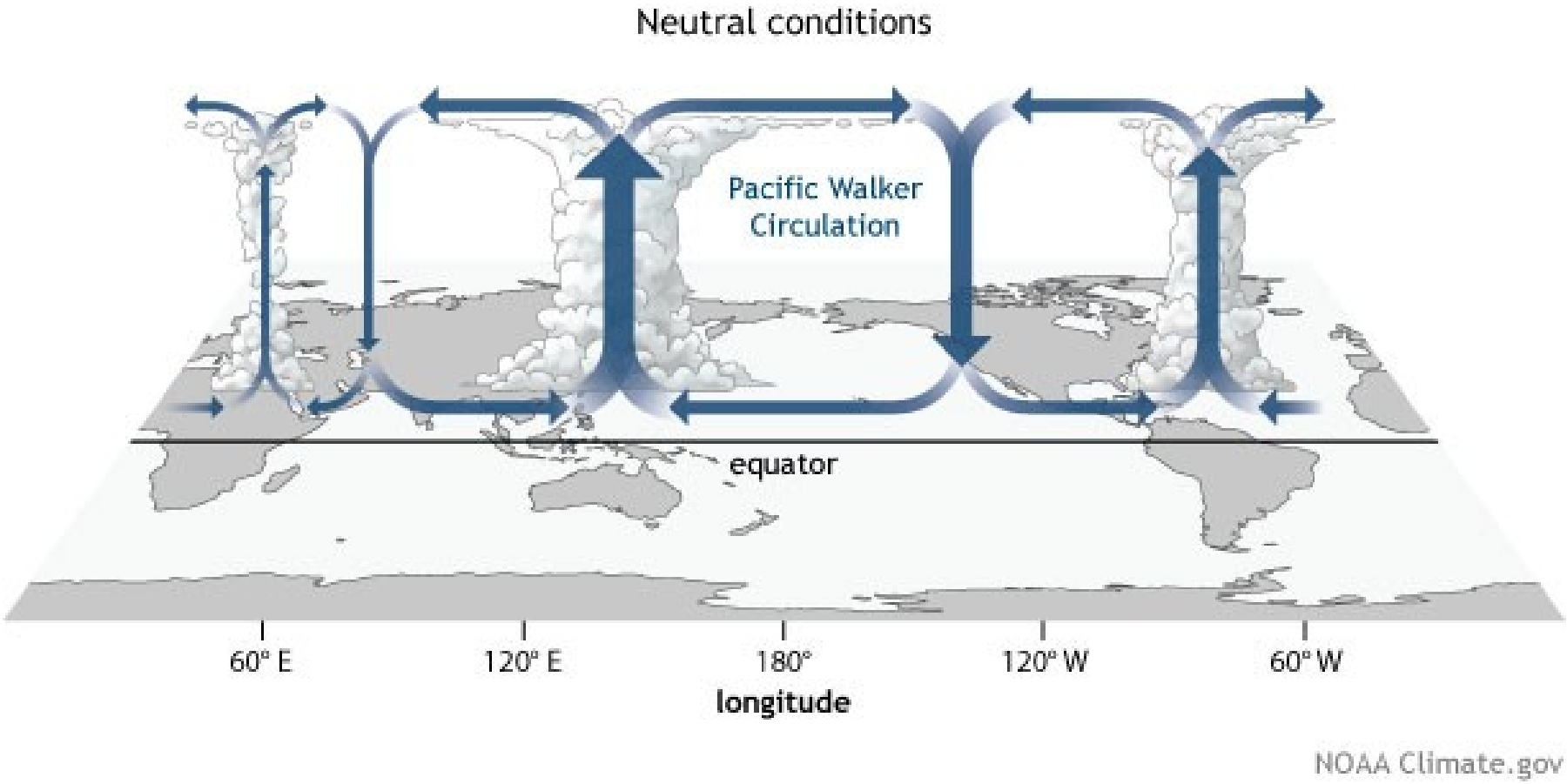


GLOBAL: 2027 a strong candidate for warmest year on record · CO_2 growth well above trend · widespread coral bleaching risk · correlated multi-breadbasket crop risk

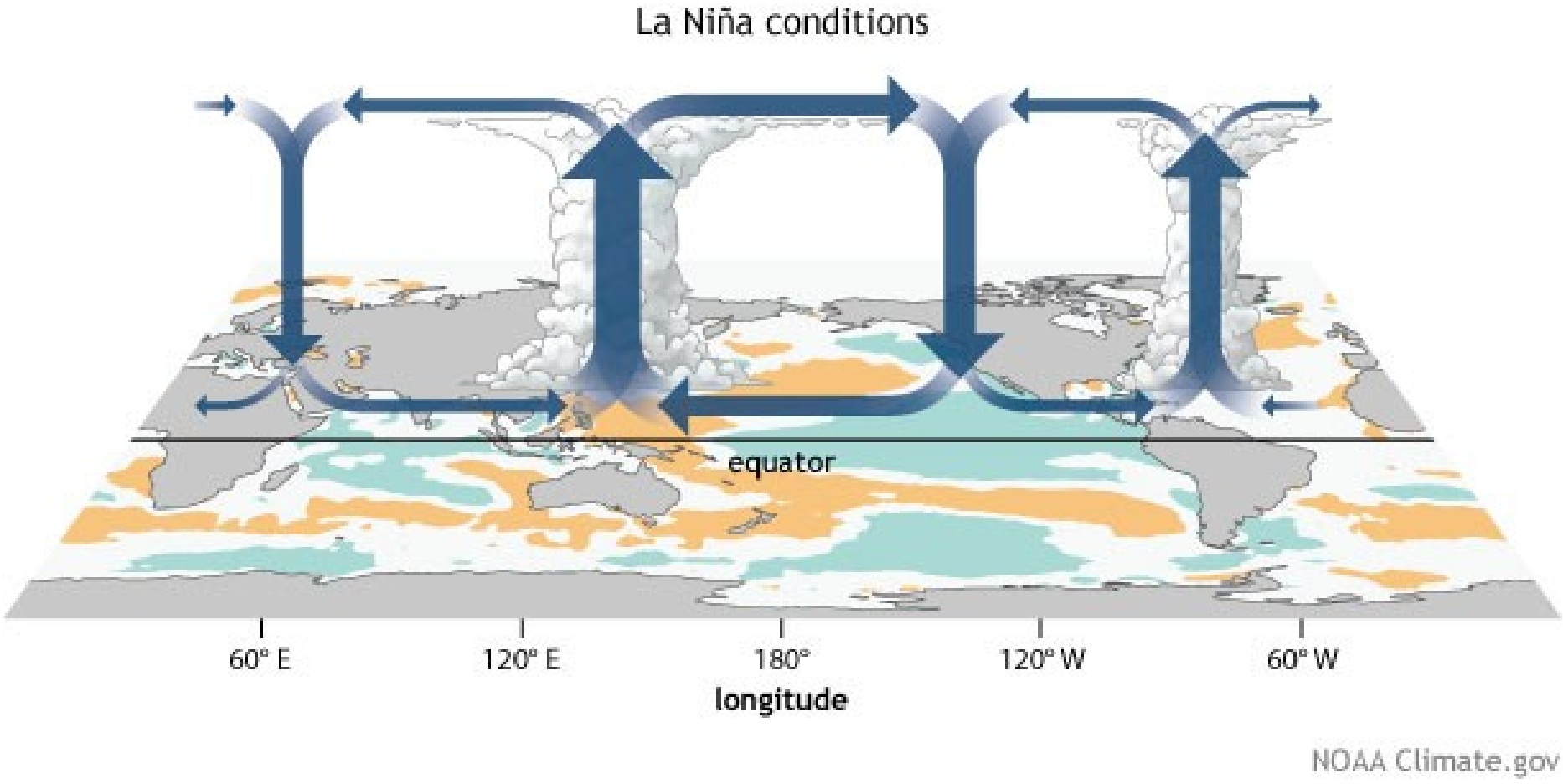
Classic ENSO



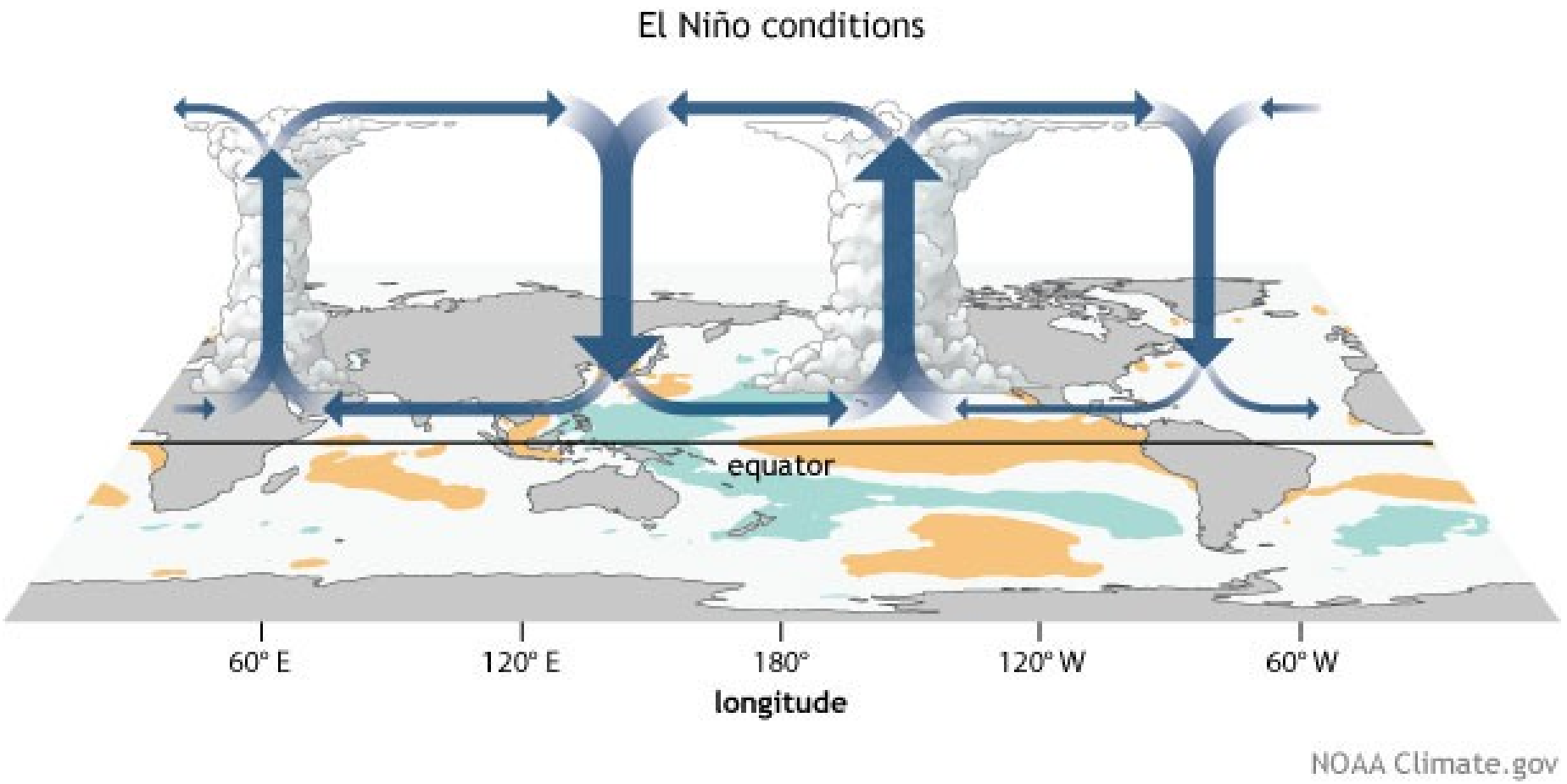
https://www.climate.gov/news-features/understanding-climate/el-niño-and-la-niña-frequently-asked-questions



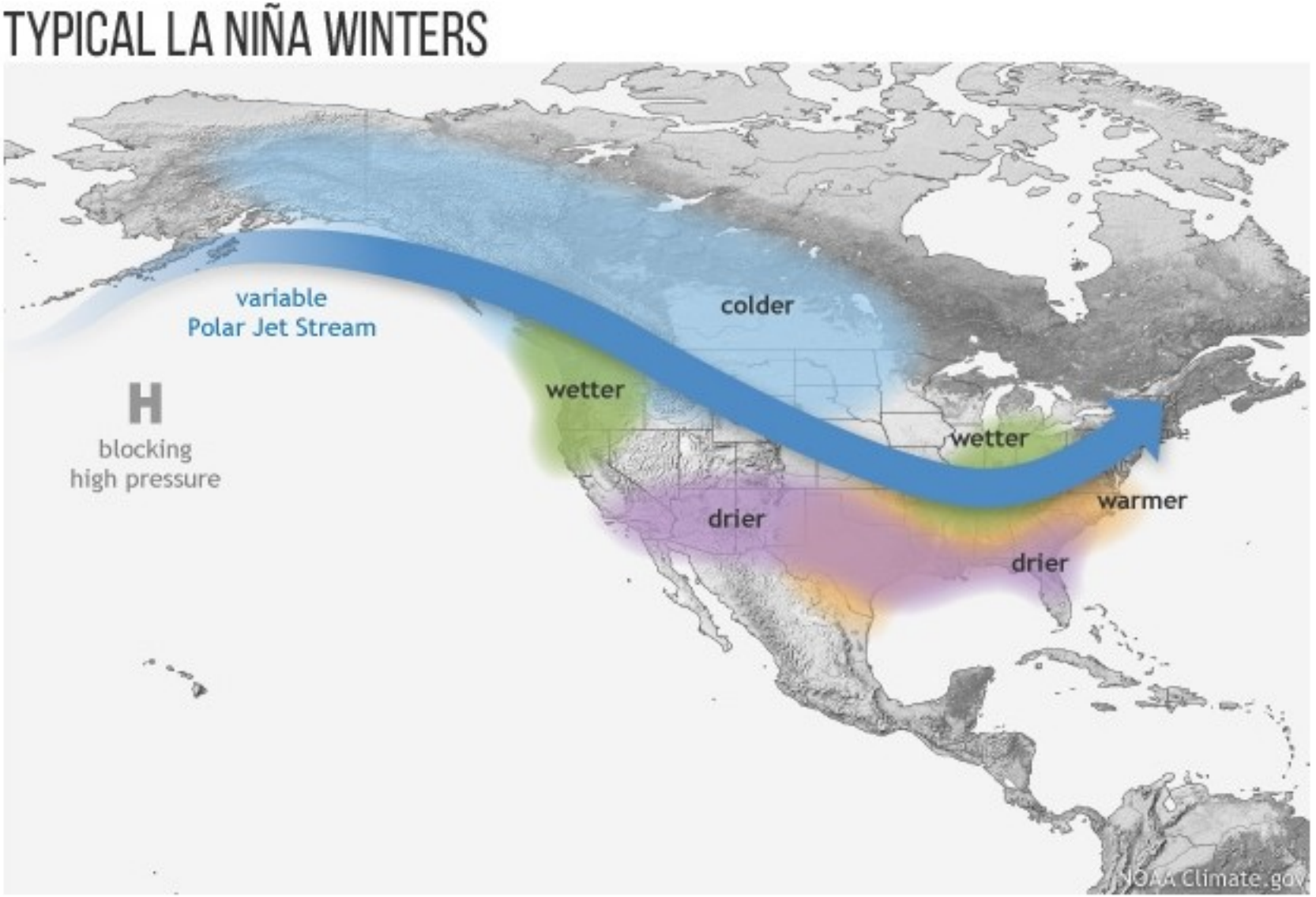
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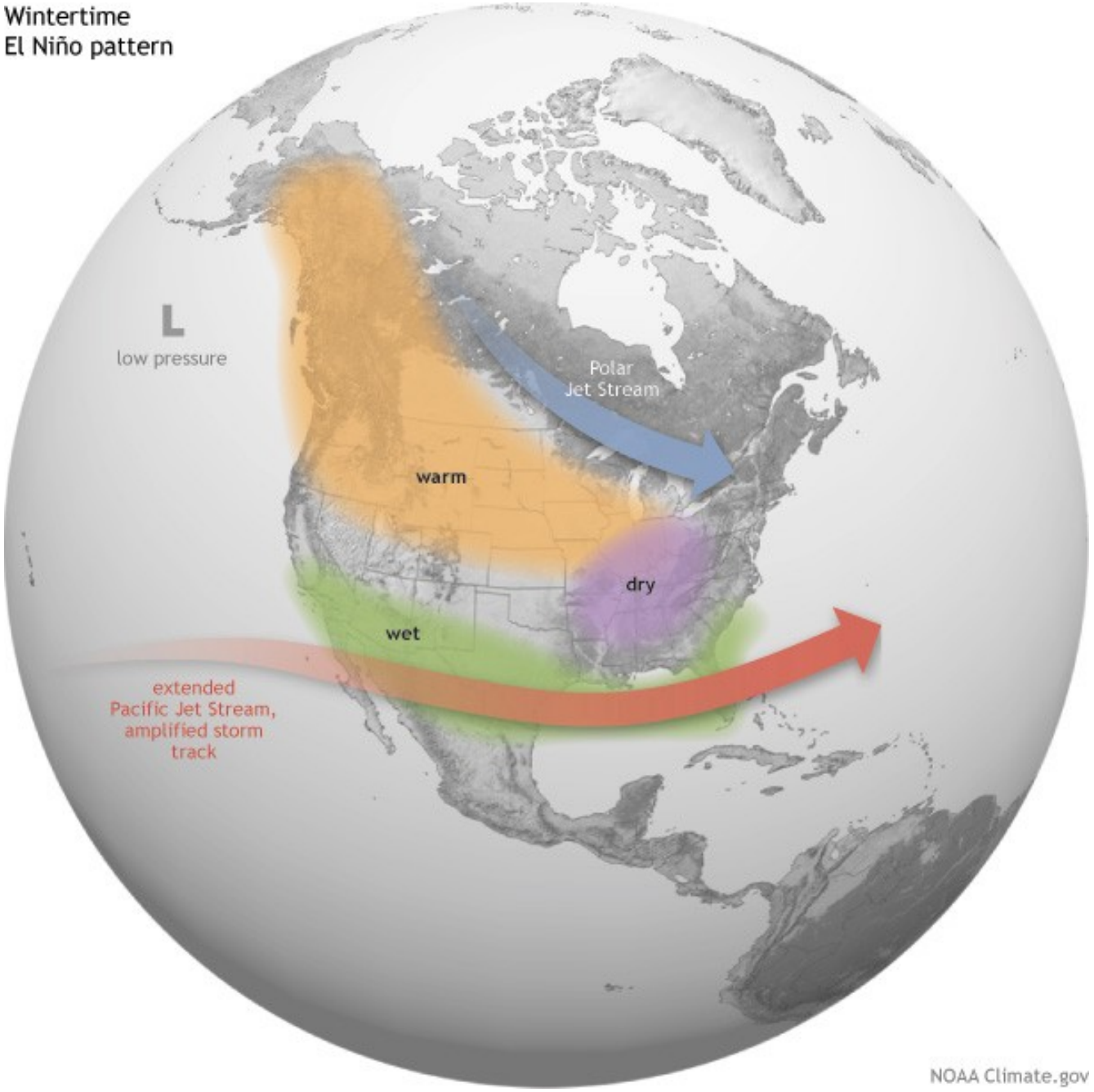
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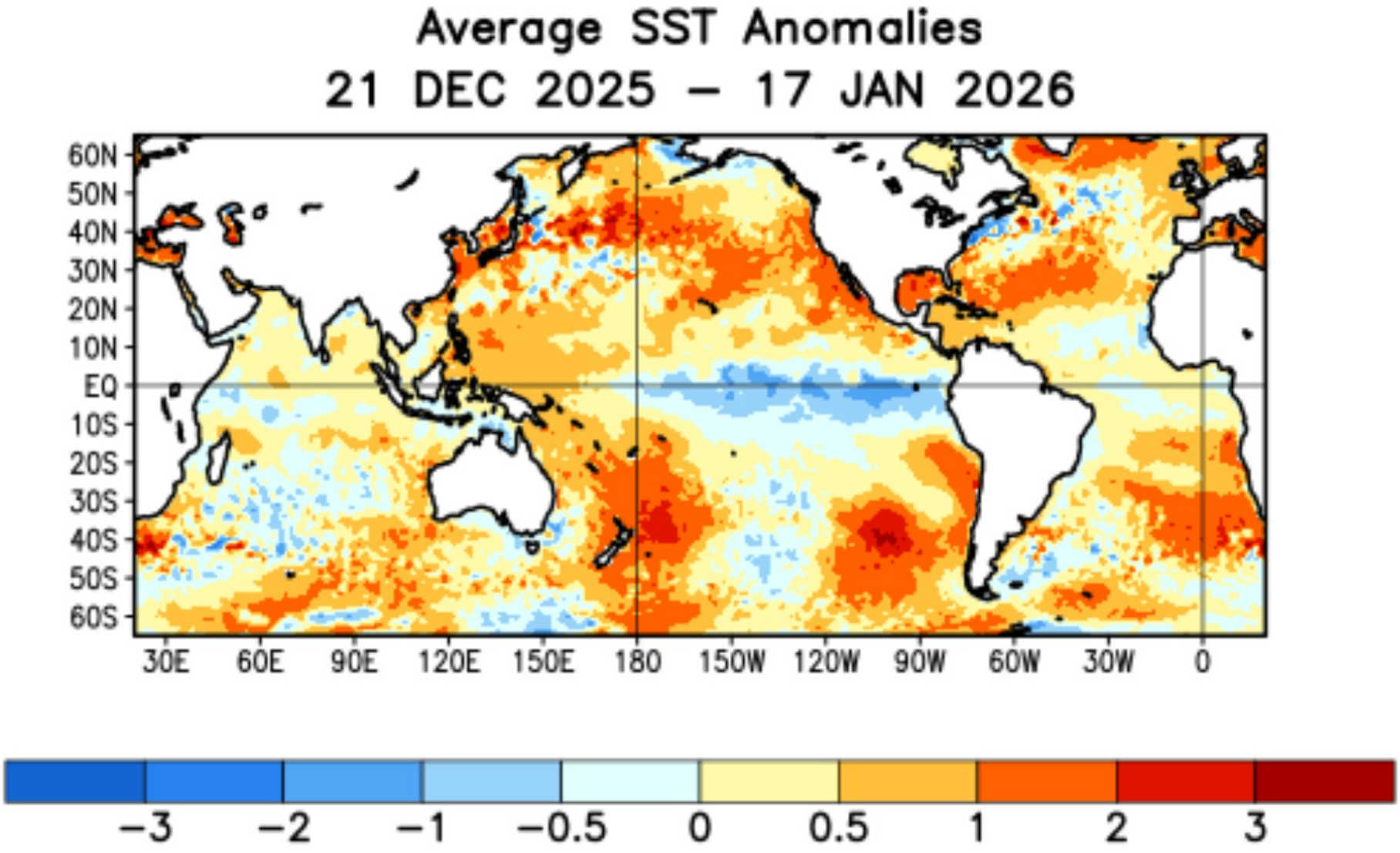


low snowpack in the Sierra Nevada of California

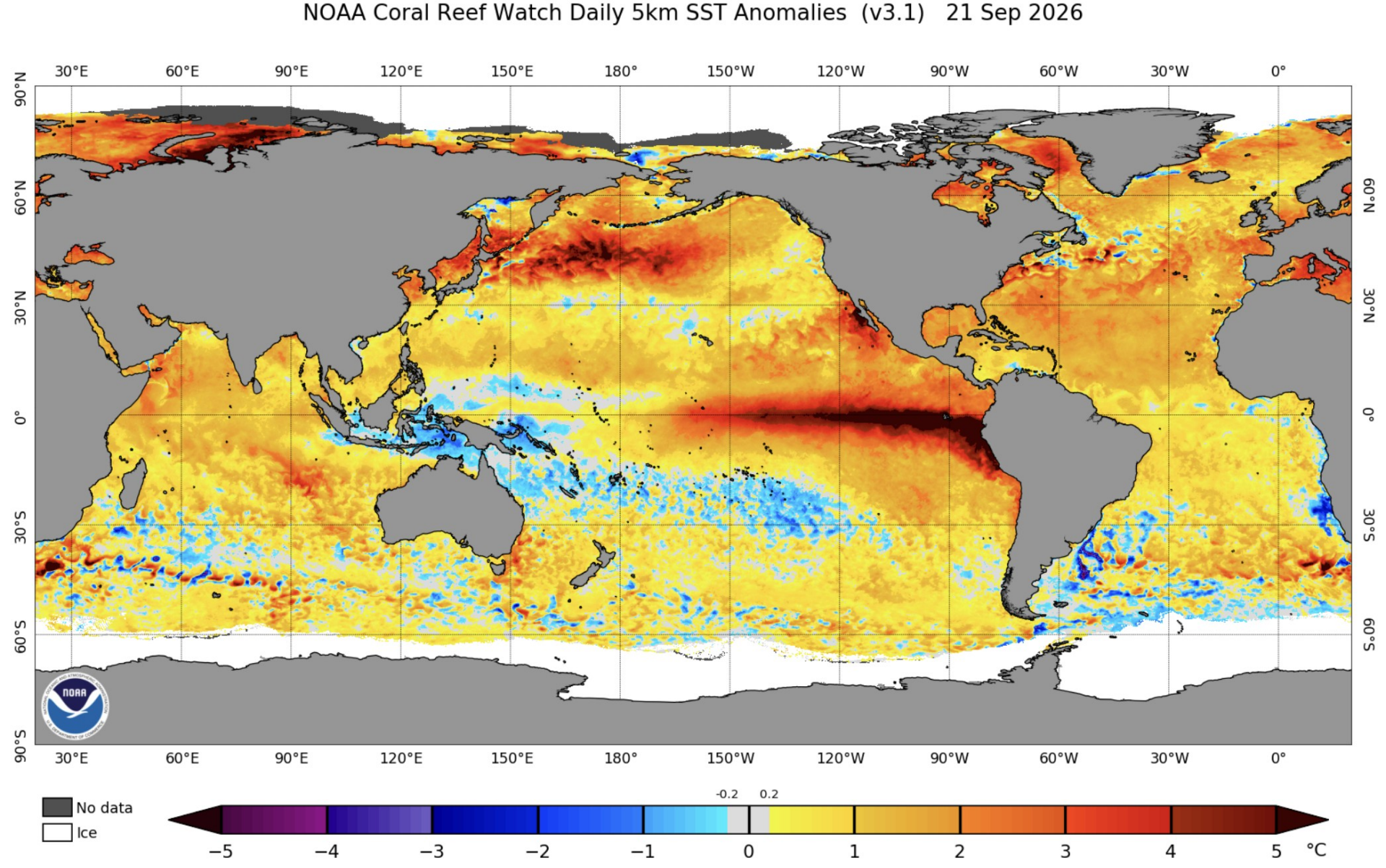


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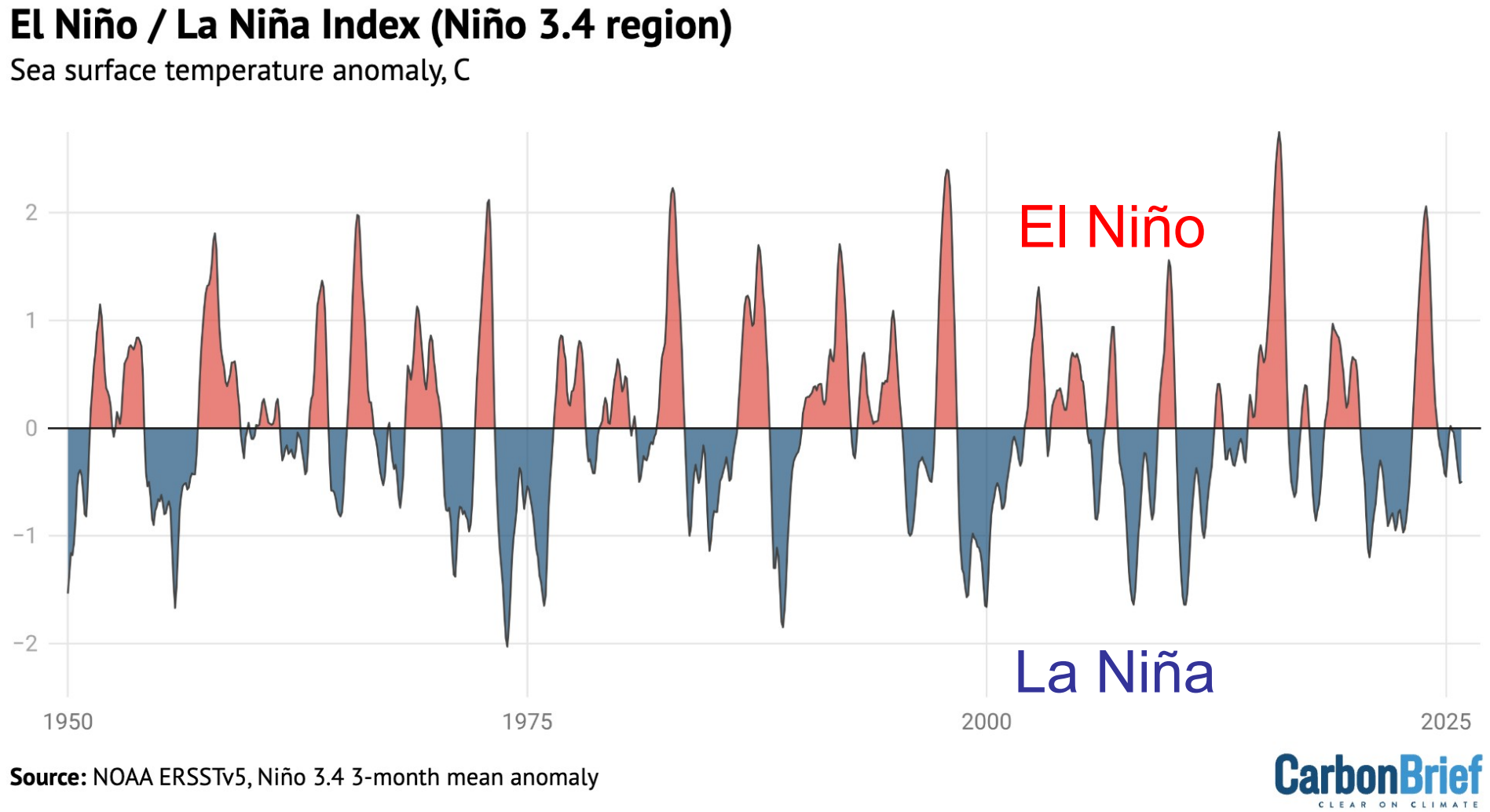
ENSO

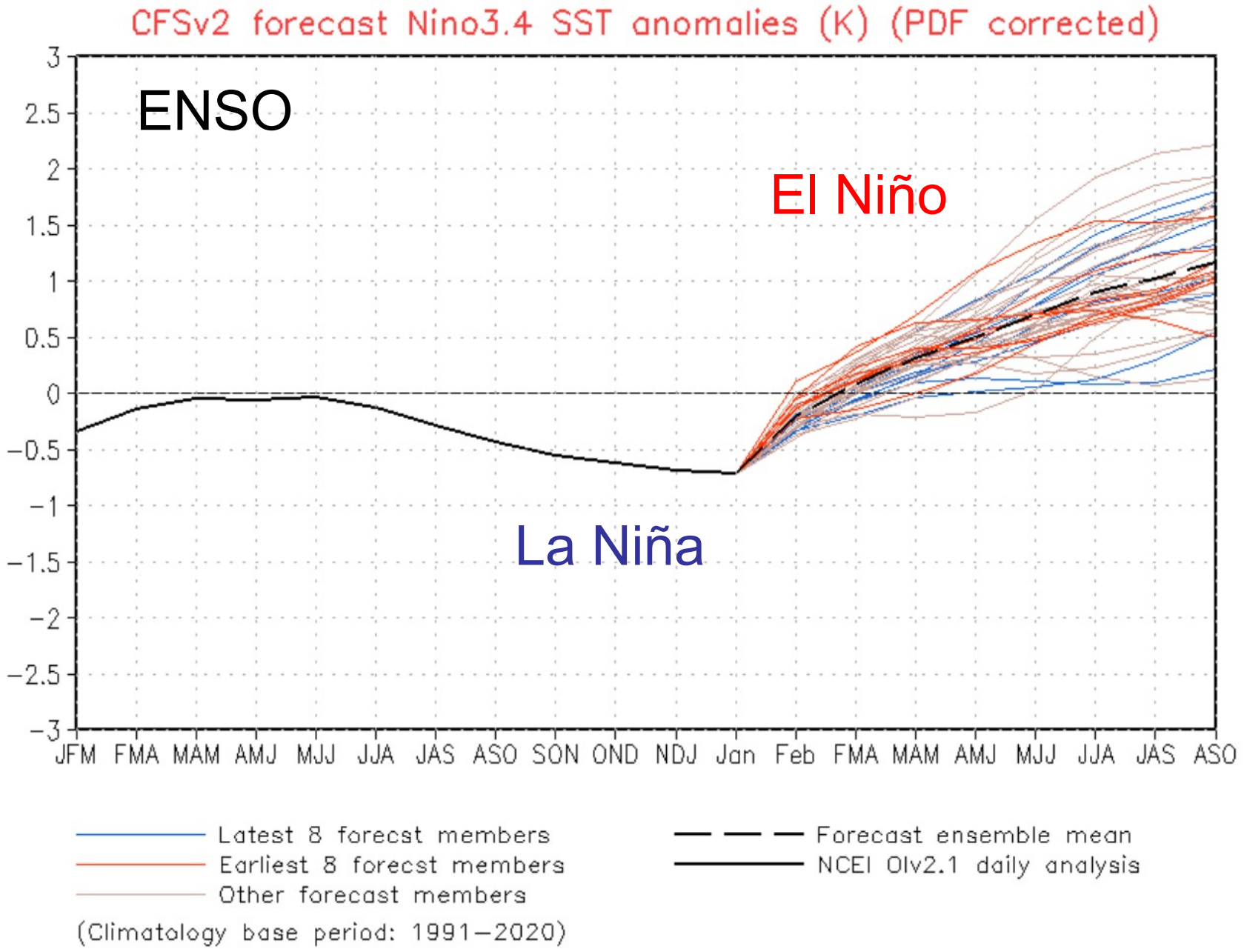

Average SST Anomalies
21 DEC 2025 – 17 JAN 2026
60N
50N
40N
30N
20N
10N
EQ
10S
20S
30S
40S
50S
60S
30E
60E
90E
120E
150E
180
150W
120W
90W
60W
30W
0
−3
−2
−1
−0.5
0
0.5
1
2
3

Current Operational SST Anomaly Charts



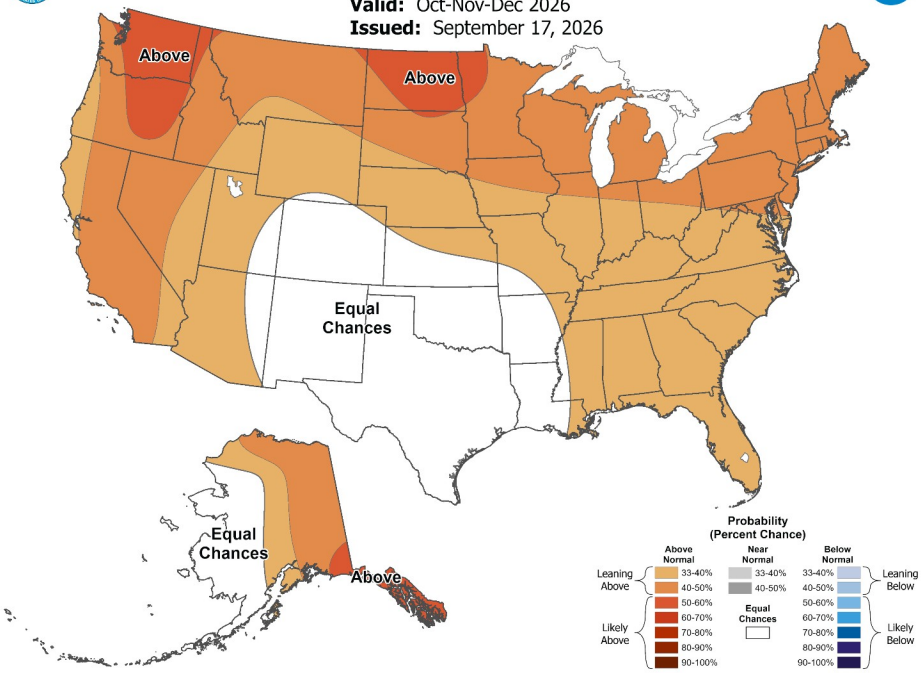
ENSO


El Niño / La Niña Index (Niño 3.4 region)
Sea surface temperature anomaly, C
2
1
0
−1
−2
1950
1975
2000
2025
El Niño
La Niña
Source: NOAA ERSSTv5, Niño 3.4 3-month mean anomaly
CarbonBrief
CLEAR ON CLIMATE


CFSv2 forecast Nino3.4 SST anomalies (K) (PDF corrected)
ENSO
El Niño
La Niña
3
2.5
2
1.5
1
0.5
0
−0.5
−1
−1.5
−2
−2.5
−3
JFM FMA MAM AMJ MJJ JJA JAS ASO SON OND NDJ Jan Feb FMA MAM AMJ MJJ JJA JAS ASO
Latest 8 forecst members
Earliest 8 forecst members
Other forecast members
Forecast ensemble mean
NCEI OIv2.1 daily analysis
(Climatology base period: 1991−2020)

Seasonal Temperature Outlook

Valid: Oct-Nov-Dec 2026
Issued: September 17, 2026



Seasonal Precipitation Outlook

Valid: Oct-Nov-Dec 2026
Issued: September 17, 2026

